

"Do what you can, with what you have, where you are."

— Theodore Roosevelt

To my family and friends.

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1 Introduction

Generating samples from a probability distribution is a central problem in statistics and an important component of modern generative modelling. In some settings, the target distribution is specified through a density known only up to a normalising constant. In others, the distribution itself is unknown and only a dataset of observations is available. These two settings lead to different sampling methods, but share a common question: how close is the distribution of the generated samples to the intended target?

This question becomes particularly challenging in high dimensions. Direct sampling is often unavailable, and the geometry of the target distribution can make it difficult for an algorithm to explore regions carrying substantial probability mass. Stochastic processes provide a useful framework for addressing these difficulties. By designing suitable dynamics and studying the evolution of their marginal distributions, we can both construct sampling algorithms and investigate their convergence properties.

A classical example is Langevin diffusion. Under suitable assumptions, its dynamics admit the target distribution as an invariant measure, and the law of the process approaches that measure over time. This construction connects sampling to the theory of Markov processes, infinitesimal generators, and functional inequalities. In particular, Poincaré and logarithmic Sobolev inequalities provide tools for quantifying convergence in appropriate distances or divergences. Their role is not merely qualitative: the constants appearing in these inequalities influence the resulting convergence rates.

Continuous-time dynamics, however, do not immediately yield an implementable algorithm. In general, their simulation requires a numerical approximation. Discretising Langevin diffusion produces an algorithm that uses evaluations of the gradient of the log-density together with Gaussian noise. This introduces a trade-off: the algorithm must run long enough to reduce the effect of its initial distribution, while its step size must be chosen to control the error caused by discretisation. Acceptance–rejection corrections, alternative discretisation schemes, and changes to the underlying dynamics offer different ways of managing this trade-off.

Theoretical guarantees make these choices more explicit. A non-asymptotic bound describes the sampling error after a finite number of iterations and shows how the required computational effort depends on quantities such as the dimension, the desired

accuracy, and the regularity of the target distribution. Strong log-concavity provides a setting in which these dependencies can be studied particularly clearly. More general targets require different assumptions and techniques; in particular, guarantees established for strongly log-concave distributions cannot be transferred automatically to multimodal distributions.

Diffusion models approach sample generation from a different starting point. Rather than assuming access to the target density or its gradient, they use observations from the target distribution to learn a sampling procedure. A forward process gradually perturbs the data with noise, producing distributions that become easier to sample from. Generation then proceeds through an approximation of the corresponding reverse-time dynamics. In the score-based formulation, these reverse dynamics depend on the score function, the gradient of the log-density of the perturbed data distribution. Since this function is generally unknown, it is estimated from data, typically using a neural network [28].

This construction introduces several sources of error. The learned score approximates an unknown function; the reverse process is usually initialised from a tractable distribution that only approximates the terminal law of the forward process; and its dynamics must be simulated numerically. Convergence analysis aims to explain how these errors affect the distribution of the generated samples. Such guarantees must be interpreted together with their assumptions. In particular, a bound conditional on an accurate score estimate does not, by itself, establish the amount of data or computation needed to learn that estimate.

The aim of this thesis is to review and explain selected theoretical convergence guarantees for the stochastic processes and algorithms underlying these sampling methods. The emphasis is on the connection between probabilistic dynamics, analytical assumptions, and finite-time error bounds. Rather than providing an exhaustive survey, the thesis develops the mathematical tools needed to understand representative results for classical sampling algorithms and diffusion models, highlighting both their implications and their limitations.

The thesis is organised as follows. Section 2 introduces the Markov-process framework, including transition semigroups, infinitesimal generators, invariant measures, and the functional inequalities used to study convergence. It then specialises these ideas

to diffusion processes and Langevin dynamics. Section 3 examines sampling from log-concave distributions, with particular attention to Langevin Monte Carlo, Metropolis–Hastings corrections, and alternative computational methods. Section 4 presents diffusion models and discusses convergence guarantees that account for score-estimation error, approximate initialisation, and time discretisation. Section 5 summarises the main conclusions and outlines open questions. The appendix collects supporting material on convexity, spectral theory, and stochastic calculus.

2 Markov Processes

This chapter develops the Markov-process framework used to study convergence to equilibrium. We first introduce transition semigroups and their generators, then relate functional inequalities to quantitative convergence bounds. Finally, we specialise these results to diffusion processes and overdamped Langevin dynamics. The presentation follows [17], [2] and [25].

A stochastic process can be described through its finite-dimensional distributions or through a probability measure on a suitable path space. The Kolmogorov extension theorem constructs a measure on the coordinate path space from consistent finite-dimensional distributions. Additional assumptions are required to obtain continuous or càdlàg paths. Wiener measure is the law of Brownian motion on continuous-path space.

2.1 Theoretical Formulation

Let's assume there exists a Markov Process $(X_t)_{t \geq 0}$ defined on an underlying filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F})_{t \geq 0}, \mathbb{P})$, taking values on a measurable space $(E, \mathcal{E})$. Then, a *Markovian transition kernel* is a map $P : E \times \mathcal{E} \rightarrow [0, 1]$ satisfying the following two properties:

- (i) $\forall x \in E$, the mapping $\mathcal{E} \ni A \mapsto P(x, A)$ is a probability measure on $(E, \mathcal{E})$,
- (ii) $\forall A \in \mathcal{E}$, the mapping $E \ni x \mapsto P(x, A)$ is $\mathcal{E}$ -measurable.

This object, i.e. the Markovian transition kernel, is a generalization of the transition matrices of discrete-state Markov Chains, and therefore it evolves the distribution of the process over time given that $\pi_t(A) = \int_E P_t(x, A) \pi_0(dx)$. We consider time-homogeneous processes: their transitions depend on the elapsed time, rather than on the starting time.

Definition 1 (Transition semigroup). *A family of Markov kernels $\mathbf{P} = (P_t)_{t \geq 0}$ is a transition semigroup if*

- (i) $\forall x \in E, P_0(x, dy) = \delta_x(dy)$ meaning P_0 is the identity operator,
- (ii) $\forall s, t \geq 0, A \in \mathcal{E}$,

$$P_{t+s}(x, A) = \int_E P_s(x, dy) P_t(y, A),$$

meaning $P_{s+t} = P_s \circ P_t$ (it is a convolution), which is known as the Chapman-Kolmogorov identity,

(iii) $\forall A \in \mathcal{E}$, the mapping $(t, x) \mapsto P_t(x, A)$ is measurable with respect to the product sigma-algebra $\mathcal{B}(\mathbb{R}_+) \otimes \mathcal{E}$.

Then, given $f \in B(E) = \{f : E \rightarrow \mathbb{R} \mid f \text{ is bounded and measurable}\}$, where $B(E)$ is endowed with the sup-norm, $\|f\|_\infty = \sup_{x \in E} |f(x)|$, we have that the same notation P_t denotes the operator

$$\begin{aligned} P_t f(x) &:= \mathbb{E}[f(X_t) | X_0 = x], \\ &= \int_E f(y) P_t(x, \mathbf{d}y), \\ &= \int_E f(y) p_t(x, y) \mathbf{d}y, \end{aligned}$$

where $p_t(x, y)$ is the density kernel of the probability measure $P_t(x, \mathbf{d}y)$ with respect to the Lebesgue measure, assuming $P_t(x, \mathbf{d}y) \ll \mathbf{d}y \quad \forall t \geq 0$. Furthermore, the map $B(E) \ni f \mapsto P_t f$ is a contraction of $B(E)$, that is $\|P_t f\|_\infty \leq \|f\|_\infty$ (notice that such a contraction property can usually be extended to different norms, e.g. $\mathbb{L}^p$ -norms. In fact, in the sequel we will work with the Hilbert space $\mathbb{L}^2$ whose squared-norm is $\|f\|_{\mathbb{L}^2(m)}^2 = \int_E f(x)^2 m(\mathbf{d}x)$, for some proper measure m on E). It is clear from property (i) of Definition [1], that we have $\lim_{t \downarrow 0} P_t f(x) = f(x)$. Notice that, if we choose f to be the indicator function, that is $f = \mathbb{1}_A$ with $A \in \mathcal{E}$, the expression of $P_t f(x)$ becomes $P(X_{s+t} \in A | X_r, 0 \leq r \leq s) = P_t(X_s, A)$, highlighting the so called *Markov property*. Consequently, we can show that the initial law and the kernels determine all finite-dimensional distributions; if we let the initial distribution of the process be $\pi_0 := \text{law}(X_0)$, then if we take $0 < t_1 < t_2 < \dots < t_p$ and $A_0, \dots, A_p \in \mathcal{E}$, $p \in \mathbb{N}$, we have,

$$\begin{aligned} P(X_0 \in A_0, X_{t_1} \in A_1, \dots, X_{t_p} \in A_p) &= \int_{A_0} \pi_0(\mathbf{d}x_0) \int_{A_1} P_{t_1}(x_0, \mathbf{d}x_1) \times \\ &\quad \times \int_{A_2} \dots \int_{A_p} P_{t_p - t_{p-1}}(x_{p-1}, \mathbf{d}x_p), \end{aligned}$$

and this also applies to other functions satisfying the same regularity conditions.

Therefore, a Markov process $(X_t)_{t \geq 0}$ with transition semigroup $\mathbf{P}$ is an $(\mathcal{F}_t)$ -adapted process with values in E such that for each $s, t \geq 0$ s.t. $s < t$ and $f \in B(E)$,

$\mathbb{E}[f(X_{s+t})|\mathcal{F}_s] = P_t f(X_s)$. However, given a semigroup, does a Markov process with that semigroup exist? Is it unique? The first step to prove this is to state a specific version of the famous *Kolmogorov extension theorem*. First, let's define the sample space, which consists, as custom, in a functional space, i.e. $\Omega^* = E^{\mathbb{R}_+}$; thus, $w : \mathbb{R}_+ \rightarrow E$, $w \in \Omega^*$. Then, we equip the sample space with the σ -algebra $\mathcal{F}^*$ generated by coordinate evaluations and for which the coordinate mappings $w \mapsto w(t)$ are measurable for $t \in \mathbb{R}_+$. With $C(\mathbb{R}_+)$ we denote the collection of all finite subsets of the positive real line, and for every $U \in C(\mathbb{R}_+)$, let $w_U : \Omega^* \rightarrow E^U$ be the mapping associating to each path w its restriction to U . At the same time, given $U, V \in C(\mathbb{R}_+)$ with $U \subset V$, then we can write $w_U^V : E^V \rightarrow E^U$.

Theorem 1 (Kolmogorov extension theorem). *Let E be a Polish space equipped with its Borel σ -algebra $\mathcal{E}$. For every $U \in C(\mathbb{R}_+)$, let P_U be a probability measure on E^U . Then, assume that the collection $(P_U, U \in C(\mathbb{R}_+))$ is consistent in the following sense: given $U, V \in C(\mathbb{R}_+)$, such that $U \subset V$, then P_U is the pushforward of P_V under w_U^V , that is $(w_U^V)_\# P_V = P_U$. If this holds true then there exists a unique probability measure P on $(\Omega^*, \mathcal{F}^*)$ such that $(w_U)_\# P = P_U$ for every $U \in C(\mathbb{R}_+)$.*

It should be clear that the theorem states that, given a collection of finite-dimensional distributions, satisfying the consistency condition, then we can construct a probability measure of a random process having those exact finite-dimensional marginals. Consequently, we can specialize the preceding theorem to Markov processes.

Corollary 1. *Assume that E is Polish and that $(P_t)_{t \geq 0}$ is a transition semigroup on E . Let π_0 be a probability measure on E . Then there exists a unique probability measure P on Ω^* under which the canonical process $(X_t)_{t \geq 0}$ is a Markov process with transition semigroup $(P_t)_{t \geq 0}$ and $X_0 \sim \pi_0$.*

Notice that the adjective 'canonical' refers to the construction of the process, where $X_t(w) = w(t)$, meaning that the value of the process at time t is just the evaluation, at time t , of the trajectory function. Hence, we showed that given a specific Markovian semigroup, there exists a unique Markov process associated to it.

For the next analytical construction, assume that E is locally compact, separable, and metrizable. A *Feller semigroup* is a Markov semigroup that maps $C_0(E)$ into itself and is

strongly continuous there:

$$\lim_{t \downarrow 0} \|P_t f - f\|_\infty = 0, \quad f \in C_0(E).$$

Here $C_0(E)$ consists of continuous functions vanishing at infinity. Under the standard Feller-process construction these assumptions yield a càdlàg strong Markov realisation. Feller regularity is a sufficient framework for this construction, not a necessary property of every strong Markov process. Strongly continuous semigroups on other function spaces also have generators.

Definition 2 (Infinitesimal generator). *For a strongly continuous semigroup $(P_t)_{t \geq 0}$ on $C_0(E)$, the infinitesimal generator $\mathcal{L}$ is defined via,*

$$\mathcal{L}f := \lim_{t \downarrow 0} \frac{P_t f - f}{t}, \quad (2.1)$$

with convergence in the supremum norm. Its domain $\mathcal{D}(\mathcal{L})$ consists of all functions for which this limit exists in $C_0(E)$.

The set containing all such functions is the domain of the generator, that we denote as $\mathcal{D}(\mathcal{L})$. For what concerns here, the infinitesimal generator uniquely identifies the Markov semigroup and therefore the Markov process (a probability measure on the space of either right-continuous or continuous functions having values in E). For $f \in \mathcal{D}(\mathcal{L})$, $P_t f \in \mathcal{D}(\mathcal{L})$ and

$$P_t f = f + \int_0^t P_s(\mathcal{L}f) ds = f + \int_0^t \mathcal{L}(P_s f) ds, \quad (2.2)$$

where the integrals are understood in the Banach space. Consequently, $\forall x \in E$, the map $t \mapsto P_t f(x)$ is continuously differentiable on $\mathbb{R}_+$ and satisfies the differential equation

$$\frac{d}{dt} P_t f = \mathcal{L} P_t f = P_t \mathcal{L} f, \quad P_0 f = f. \quad (2.3)$$

This is the abstract Kolmogorov backward equation. The solution is $P_t = e^{t\mathcal{L}}$, highlighting the positivity property of $\mathbf{P}$; for an unbounded generator it need not be defined by an everywhere convergent operator power series.

The dual action on measures is characterised by

$$\int_E f \, d(P_t^* \mu) = \int_E P_t f \, d\mu.$$

Thus $\pi_t = P_t^* \pi_0$, and the forward equation is, at least weakly,

$$\frac{d}{dt} \int_E f \, d\pi_t = \int_E \mathcal{L} f \, d\pi_t, \quad f \in \mathcal{D}(\mathcal{L}). \quad (2.4)$$

Writing $\partial_t \pi_t = \mathcal{L}^* \pi_t$ is shorthand for this dual equation. It should be distinguished from a Hilbert-space adjoint acting on densities. When a density exists relative to a reference measure m , its evolution involves the adjoint relative to that particular m .

When working with a Markov semigroup $(P_t)_{t \geq 0}$ with infinitesimal generator $\mathcal{L}$ and stationary distribution π , the usual function space taken into consideration is the Hilbert space of real-valued functions which are measurable and square-integrable with respect to the invariant measure, namely $\mathbb{L}^2(\pi)$. Then, an important and easier to study setting is identified by a symmetric Markov semigroup, as given by the following definition.

Definition 3. *The Markov semigroup $(P_t)_{t \geq 0}$ is symmetric, therefore reversible, with respect to the invariant measure π if for all $f, g \in \mathbb{L}^2(\pi)$ and $t \geq 0$ we have,*

$$\langle P_t f, g \rangle_{\mathbb{L}^2(\pi)} = \langle f, P_t g \rangle_{\mathbb{L}^2(\pi)} \quad (2.5)$$

The above equality assumes that the process is already at stationarity, namely $X_0 \sim \pi$. A Markov Process being π -reversible is π -stationary and time-reversible; in fact, if we take, for example, two indicator functions, $f = \mathbb{1}_A$ and $g = \mathbb{1}_B$ for measurable sets A, B , then

$$\begin{aligned} \int_E (P_t f) g \, d\pi &= \int_B P(X_t \in A | X_0 \in B) \, d\pi, \\ &= P(X_t \in A, X_0 \in B), \\ &= P(X_0 \in A, X_t \in B), \\ &= \int_E f (P_t g) \, d\pi. \end{aligned}$$

When working within the Hilbert space $\mathbb{L}^2(\pi)$ we have that the semigroup as well as its infinitesimal generator are self-adjoint (consequently, they have real spectrum), i.e.

$P_t = P_t^*$ and $\mathcal{L} = \mathcal{L}^*$. Then it is natural to consider the density $\rho_t := \pi_t/\pi$ as it satisfies the Fokker-Planck equation, $\partial_t \rho_t = \mathcal{L} \rho_t$. Then the process will converge, with some velocity, to the stationary measure π , resulting in $\partial_t \rho_t = 0$.

But how fast does the process converge to stationarity? Well, this is quite an intriguing question whose answer is kind of intricate due to its dependence on several factors, such as the geometry of the space as well as the shape of the invariant distribution, e.g. it is easier to converge to a unimodal and strong log-concave distribution such as a Gaussian than to a multimodal distribution. A process, in order to achieve stationarity, must mix well, in the sense that it needs to explore the state space as fast as possible, taking into account the relative 'importance' of regions of the distribution. In mathematical terms, we might define the mixing time as the first time that the distribution of the process is ϵ close to the invariant measure, that is, $t_{\text{mix}}(\epsilon, \pi_0) = \inf\{t \in \mathbb{R} | d_{TV}(\pi_t, \pi) \leq \epsilon\}$, where TV stands for Total Variation, a distance used for probability measures that will be introduced later on.

We are interested in the behavior of $P_t f$ as $t \rightarrow \infty$. We know that the spectral decomposition of $P_t f$ is given by $P_t f = \int_0^\infty e^{-\lambda t} dE_\lambda f$, for any $f \in \mathbb{L}^2(\pi)$. From this we get that $\lim_{t \rightarrow \infty} P_t f = 0$, $\forall \lambda > 0$, namely $P_t f$ converges in $\mathbb{L}^2(\pi)$ to the projection of f on the space of functions satisfying $\mathcal{L} f = 0$, that is to the kernel of the infinitesimal generator. The spectral components with $\lambda > 0$ do not matter anymore, because they vanish exponentially as $t \rightarrow \infty$. Hence, we have that, in terms of spectral decomposition, the functions f such that $\mathcal{L} f = 0$, are associated with $\lambda = 0$, meaning that $\lim_{t \rightarrow \infty} P_t f = dE_0 f$, which corresponds to the projection of f onto $\text{Ker}(\mathcal{L})$. In order to be able to formally state the previous result, we need to define the concept/property of *ergodicity*, which implies that a process is irreducible, positive recurrent, and non-explosive.

Definition 4. *The infinitesimal generator $\mathcal{L}$ is said to be ergodic if every $f \in \mathcal{D}(\mathcal{L})$ such that $\mathcal{L} f = 0$ is constant. In particular, $f = 0$ if π has infinite mass (e.g. Lebesgue).*

The further requirement for the case where π has infinite mass is due to the fact that $\mathcal{D}(\mathcal{L}) \subset \mathbb{L}^2(\pi)$, so that we need $\int_E f^2 d\pi < \infty$; when π has infinite mass over E the only way for f to satisfy the above constraint is to be constantly equal to 0. Then, another important property that the semigroup must satisfy is the *mass conservation* one. This is stated in the following definition.

Definition 5. The semigroup $(P_t)_{t \geq 0}$ is said to be conservative if for every $t \geq 0$ we have $P_t(\mathbb{1}) = \mathbb{1}$.

This property is crucial to have a well-defined Markov process. Combining the last two notions, we obtain the following proposition, as in [2].

Proposition 1. Assume that ergodicity holds. Then it holds:

- If $\pi(E) = \infty$ then, for any $f \in \mathbb{L}^2(\pi)$, $\lim_{t \rightarrow \infty} P_t f = 0$ in $\mathbb{L}^2(\pi)$.
- If $\pi(E) < \infty$ and the semigroup is conservative, for any $f \in \mathbb{L}^2(\pi)$,

$$\lim_{t \rightarrow \infty} P_t f = \int_E f d\pi, \text{ in } \mathbb{L}^2(\pi).$$

An example of a process having $\pi(E) < \infty$ with a conservative semigroup is the Ornstein-Uhlenbeck process, that will play an important role in the diffusion models setting. Its SDE formulation is,

$$dX_t = \theta(x^* - X_t)dt + \sigma dB_t,$$

we have the deterministic mean reversion component, i.e. $\theta(x^* - X_t)dt$, which pushes the process to its stationary distribution, namely the d-dimensional Gaussian having x^* as mean and $\sigma^2/2\theta$ as variance.

Finally, let us introduce the *Dirichlet energy/form* whose concept is key in defining Markov processes due to the fact that usually we start by defining this bilinear functional and then, by also using the Riesz Representation Theorem, we prove the existence of the corresponding infinitesimal generator. It will appear in the two functional inequalities that we are going to treat in the next sub-section.

Definition 6. Assume that the Markov semigroup $(P_t)_{t \geq 0}$ is reversible with self-adjoint generator $\mathcal{L}$ and stationary distribution π . Then, the Dirichlet energy associated with $\mathcal{L}$ is the bilinear form,

$$\mathcal{E}(f, g) := \langle f, (-\mathcal{L})g \rangle_{\mathbb{L}^2(\pi)} = \langle (-\mathcal{L})f, g \rangle_{\mathbb{L}^2(\pi)}, \quad (2.6)$$

for functions f, g in its domain, $\mathcal{D}(\mathcal{E})$.

In particular, we have $\mathcal{E}(f, f) \geq 0$, $\forall f \in \mathcal{D}(\mathcal{E})$, showing that the operator $-\mathcal{L}$ is non-negative (it has non-negative real spectrum). This functional measures the smoothness of suitable functions, weighting for the stationary measure. In fact, when treating the family of diffusion processes defined over a d -dimensional Euclidean space, we will see that such a Dirichlet form, $\mathcal{E}(f, f)$, will specialize to be the expected square gradient of the function f .

2.2 Functional Inequalities

Regarding Proposition [1], we are interested in the second framework, i.e. $\pi(E) < \infty$; thus, we notice that the conditional expected value $P_t f(x) = \mathbb{E}(f(X_t) | X_0 = x)$ converges to the unconditional (stationary) one $\mathbb{E}_\pi(f(X_t))$, because stationarity implies that the dynamics of the process do not change its marginal distribution over time. But how fast does the process convergence to its stationary distribution? Well, if the stochastic process (its stationary measure) satisfies certain functional inequalities then it will be exponentially ergodic, meaning it will converge exponentially fast to the stationary measure.

2.2.1 Poincaré inequality

The first functional inequality that we treat is the *Poincaré* or *Spectral Gap inequality*. It is the simplest, if we are allowed to say so, inequality which quantifies ergodicity, giving results about the convergence of the semigroup, $\mathbf{P} = (P_t)_{t \geq 0} \rightarrow \pi$. Given the representation $P_t f(x) = \int_E f(y) P_t(x, dy)$, $t \geq 0, x \in E$, where we can also assume the transition kernel to have a density with respect to the Lebesgue measure, convergence implies that $P_t(x, dy) \rightarrow d\pi(y)$, $x \in E$, as $t \rightarrow \infty$, meaning that the process forgets about the initial condition. Let's first state the Poincaré inequality and then discuss it.

Definition 7. A Markov Process $(X^x)_{t \geq 0}$ is said to satisfy a Poincaré - Spectral Gap inequality $P(C_{PI})$ with constant $C_{PI} > 0$, if for all smooth and compactly supported functions $f : E \rightarrow \mathbb{R}$ we have,

$$\text{Var}_\pi f \leq C_{PI} \mathcal{E}(f). \quad (2.7)$$

Then, the least constant $C_{PI} > 0$ such that the above inequality is satisfied is called *Poincaré constant*.

Suppose that f is an eigenfunction of the positive operator $-\mathcal{L}$, so that $-\mathcal{L}f = \lambda f$ for some eigenvalue $\lambda > 0$. Applying the Spectral Gap inequality to such a function shows that, if $\lambda \neq 0$ then $C_{PI}\lambda \geq 1$ or $\lambda \geq 1/C_{PI}$. In fact, the expected value of f is such that $\int_E f d\pi = -\frac{1}{\lambda} \int_E \mathcal{L}f d\pi = 0$ by invariance, giving us the following $P(C_{PI})$: $\int_E f^2 d\pi \leq C_{PI} \mathcal{E}(f) = C_{PI} \int_E f(-\mathcal{L}f) d\pi = C_{PI}\lambda \int_E f^2 d\pi$. Therefore, under $P(C_{PI})$, every non-zero eigenvalue of $-\mathcal{L}$ is greater or equal to $1/C_{PI}$. This holds true also when the spectrum is not just discrete, meaning that the spectrum of $-\mathcal{L}$ will be $\{0\} \cup [1/C_{PI}, +\infty)$, so that $P(C_{PI})$ describes a gap in its spectrum, and the higher the first $\lambda > 0$ the faster the convergence of the process, as it will be clear by reading the following theorem.

The main consequence of the Poincaré inequality is that the Markov semigroup $\mathbf{P}$ converges exponentially to equilibrium, in the $\mathbb{L}^2(\pi)$ - sense, as stated by the following theorem.

Theorem 2. *The following are equivalent.*

1. *The Markov process satisfies a Poincaré inequality with constant C_{PI} .*
2. *For all $f \in \mathbb{L}^2(\pi)$ with $\int_E f d\pi = 0$ and all $t \geq 0$,*

$$\|P_t f\|_{\mathbb{L}^2(\pi)}^2 \leq e^{-\frac{2t}{C_{PI}}} \|f\|_{\mathbb{L}^2(\pi)}^2. \quad (2.8)$$

The assumption about the stationary mean of f is just for simplicity. If we did not make this assumption, then the inequality would be,

$$\text{Var}_\pi(P_t f) \leq e^{-2t/C_{PI}} \text{Var}_\pi(f), \quad \forall f \in \mathbb{L}^2(\pi), t \geq 0,$$

which shows, even more clearly, the exponential decay in variance as the process evolves over time, given the assumption that a Poincaré inequality holds.

In order to use such results for the convergence of the entire distribution, namely of $\pi_t = \text{law}(X_t)$ to π , then we can just consider $f = \rho_0 - 1$ in the above setting, where $\rho_t := \pi_t/\pi$ and, as we already saw, $\rho_t = P_t \rho_0$ by the self-adjoint property of the semigroup. Hence, we obtain the so-called *chi-squared divergence*, which is not a metric as the name suggests; it has the following expression: $\chi^2(\pi_t|\pi) := \|\mathbf{d}\pi_t/\mathbf{d}\pi - 1\|_{\mathbb{L}^2(\pi)}^2 = \text{var}_\pi \mathbf{d}\pi_t/\mathbf{d}\pi$ if $\pi_t \ll \pi$, $\chi^2(\pi_t|\pi) := \infty$ otherwise. Then, we have the following theorem.

Theorem 3. *The following are equivalent.*

1. *The Markov process satisfies a Poincaré inequality with constant C_{PI} .*
2. *For any initial distribution $\pi_0 = \text{law}(X_0)$ and all $t \geq 0$,*

$$\chi^2(\pi_t \| \pi) \leq e^{-\frac{2t}{C_{PI}}} \chi^2(\pi_0 \| \pi).$$

Therefore, given an initial $\mathbb{L}^2$ -distance between radon-nikodym derivatives, we have that this distance decrease exponentially fast as a function of time. At the same time, the lower the Poincaré constant C_{PI} (i.e. the higher the first eigenvalue greater than zero of $-\mathcal{L}$), the faster the convergence, as can also be understood by looking at the spectral decomposition.

2.2.2 Log-Sobolev inequality

Another useful functional inequality which is frequently used is the so-called *log-Sobolev inequality (LSI)*, which contains more information than the Poincaré inequality due to the fact that it does provide an upper bound on the entropy of a density function and not on its variance. Furthermore, differently from what is required for the Poincaré inequality to hold, here the diffusion property is necessary.

Given a measure space $(E, \mathcal{E}, \nu)$ with ν possibly infinite, we define for all positive and integrable functions f such that $\int_E f |\log f| d\nu < \infty$ the entropy functional,

$$\mathcal{E}nt_\nu(f) = \int_E f \log f d\nu - \int_E f d\nu \log \left(\int_E f d\nu \right).$$

When ν is a probability measure then we have $\mathcal{E}nt_\nu(f) \geq 0$ and equality holds only if f is constant. In fact, if ν is a probability measure and f is not a density, then the above entropy functional corresponds to the Kullback-Leibler (KL) divergence between $\tilde{\nu}(dx) = \frac{f(x)}{\int_E f d\nu} \nu(dx)$ and ν . Now, we can define the log-Sobolev inequality.

Definition 8. *A Markov Process $(X^x)_{t \geq 0}$ is said to satisfy a tight log-Sobolev inequality with constant $C_{LSI} > 0$ if for all smooth and compactly supported functions $f : E \rightarrow \mathbb{R}$ we have,*

$$\mathcal{E}nt_\pi(f^2) \leq 2C_{LSI} \mathcal{E}(f). \quad (2.9)$$

Given the properties of the carré du champ operator we can rewrite the LSI as $\mathcal{E}nt_\pi(f) \leq \frac{C_{LSI}}{2} \int_E \frac{\Gamma(f)}{f} d\pi$. When the function f represents a density with respect to the probability measure π then the entropy functional of f is the KL divergence between $d\nu = f d\pi$ and π , namely,

$$\text{KL}(\nu||\pi) = \mathcal{E}nt_\pi(f) = \int_E \log f d\nu.$$

At the same time, we can define the *Fisher information* of ν with respect to π as

$$I(\nu|\pi) := \int_E \frac{\Gamma(f)}{f} d\pi,$$

which for $E = \mathbb{R}^d$ specializes to $I(\nu|\pi) = \int_{\mathbb{R}^d} f |\nabla \log f|^2 d\pi$. Therefore, a LSI ends up being the following,

$$\text{KL}(\nu||\pi) \leq \frac{C_{LSI}}{2} I(\nu|\pi).$$

Just as the Poincaré inequality implies an exponential decay in variance, the Log-Sobolev inequality implies an exponential decay of entropy, as stated by the following theorem.

Theorem 4. *The log-Sobolev inequality with constant C_{LSI} for the probability measure π is equivalent to saying that for every positive function $f \in L^1(\pi)$ such that $\int_E f |\log f| d\pi < \infty$,*

$$\mathcal{E}nt_\pi(P_t f) \leq e^{-2t/C_{LSI}} \mathcal{E}nt_\pi(f), \quad \forall t \geq 0. \quad (2.10)$$

This expression can be made clearer if we consider, once again, the KL divergence formulation. In fact, in the context of Markov processes we want to establish some kind of inequality which assures us about the speed of convergence of such a process. As we consider $d\pi_t = P_t f d\pi$, $t \geq 0$, that is $f = d\pi_0/d\pi$, the entropy decay equation becomes,

$$\text{KL}(\pi_t||\pi) \leq e^{-2t/C_{LSI}} \text{KL}(\pi_0||\pi), \quad (2.11)$$

meaning that as the time goes on the process exhibits an exponential decay of the KL divergence between the probability measure of the process at time t , π_t , and the equilibrium one, π . The fact that relative entropy decays along the process trajectories is validated by noticing that the time derivative of the relative entropy is the opposite

of the relative Fisher information; given that $I_\pi(\cdot) \geq 0$, it is always true that along the sample paths the relative entropy decreases. We can formalize what we said by stating the following proposition, known as *de Bruijn's identity*.

Proposition 2. *For every positive f which is smooth and compactly supported, it holds,*

$$\frac{d}{dt} \text{Ent}_\pi(P_t f) \leq -I_\pi(P_t f). \quad (2.12)$$

Furthermore, convergence in KL divergence implies convergence in total variation. Let's recall its definition.

Definition 9. *The total variation (TV) distance between π_t and π is given by,*

$$\|\pi_t - \pi\|_{TV} := \sup_{A \in \mathcal{F}} |\pi_t(A) - \pi(A)|, \quad (2.13)$$

which becomes $\|\pi_t - \pi\|_{TV} = \frac{1}{2} \int |\pi_t(x) - \pi(x)| dx$, assuming π_t and π are dominated by the Lebesgue measure (we use π_t and π to indicate the densities as well as the probability measure). It turns out that the TV distance satisfies the *Pinsker-Csiszár-Kullback inequality*, which reads,

$$\|\pi_t - \pi\|_{TV}^2 \leq \frac{1}{2} \text{KL}(\pi_t \| \pi). \quad (2.14)$$

Thus, if we control $\text{KL}(\pi_0 \| \pi)$, then the previous theorem implies the stronger convergence of the process toward equilibrium in total variation.

What about the properties the distributions need to satisfy in order to satisfy the two functional inequalities we talked about? Well, if the distribution is strongly log-concave then it satisfies both inequalities, as stated by the following Lemma.

Lemma 1. *Let π be a distribution on $E = \mathbb{R}^d$. Then it holds the following.*

1. *(Bakry-Emery-theorem) If π is α -strongly log-concave, then it satisfies a log-Sobolev inequality with constant $C_{LSI} \leq 1/\alpha$.*
2. *If π satisfies a log-Sobolev inequality with constant C_{LSI} , then it also satisfies a Poincaré inequality with constant $C_{PI} \leq C_{LSI}$.*

Therefore, if a distribution is α -strongly log-concave then the Markov process having it as equilibrium distribution will converge to it exponentially fast, given what we discussed about the Poincaré and log-Sobolev inequalities. However, it is important to remember that for the LSI we need the initial distribution to be dominated by the invariant measure, while for the PI we need the initial distribution to be square integrable with respect to π . An example of α -strongly log-concave measure is the Gaussian measure, where $\alpha = 1/\sigma^2$; the more concentrated the Gaussian is (the lower the variance), the higher the α , the lower the LSI and PI constants, and, finally, the faster the convergence to equilibrium.

Up to now we have presented results about the convergence of a continuous-time process with respect to two divergences and stated that convergence in relative entropy implies convergence in total variation distance; however, total variation distance does not consider the metric structure and the geometry of the space of probability measures, as it only measures the maximum possible difference in probability mass over all measurable sets without considering how far such a mass has to be moved in order to transform one into the other. Consequently, we might want to assess convergence results in some metric having the just mentioned properties. The ideal candidate is the 2-Wasserstein distance. Given the Polish space $(\mathbb{R}^d, \|\cdot\|_2)$, we denote the space of measures over the d -dimensional Euclidean space having finite second moment by $\mathcal{P}_2(\mathbb{R}^d)$. Then, for any measures $p_1, p_2 \in \mathcal{P}_2(\mathbb{R}^d)$, we let $\mathcal{C}(p_1, p_2)$ be the set of couplings between the two such that if $(X, Y) \sim q$, with $q \in \mathcal{C}(p_1, p_2)$, then $X \sim p_1$ and $Y \sim p_2$. Finally, we have the following definition.

Definition 10. Given two probability measures $p_1, p_2 \in \mathcal{P}_2(\mathbb{R}^d)$, the 2-Wasserstein distance between p_1 and p_2 is,

$$\mathcal{W}_2(p_1, p_2)^2 := \inf_{q \in \mathcal{C}(p_1, p_2)} \int_{\mathbb{R}^d \times \mathbb{R}^d} |x - y|^2 dq(x, y). \quad (2.15)$$

Due to the work of [21], we can establish the so-called *Talagrand's (T_2) inequality*, that is, a transportation inequality comparing the $\mathcal{W}_2$ -metric from optimal transport with the relative entropy, highlighting the contraction property of the $\mathcal{W}_2$ -metric along the Fokker-Planck trajectory.

Lemma 2. Otto-Villani Theorem Suppose that π is a distribution on $\mathbb{R}^d$ satisfying an LSI

with constant C_{LSI} . Then, for all distribution $\mu \in \mathcal{P}_2(\mathbb{R}^d)$,

$$\mathcal{W}_2^2(\mu, \pi) \leq 2C_{LSI} \text{KL}(\mu \parallel \pi). \quad (2.16)$$

Thus, along the diffusion path the $\mathcal{W}_2$ -metric is controlled by the relative entropy, if the stationary distribution satisfies a log-Sobolev inequality. In this way we can guarantee an exponential convergence to equilibrium also under a more structured metric.

There exists an equivalent inequality when π satisfies the weaker condition imposed by a Poincaré inequality; we just have to replace the KL divergence with the chi-squared one and C_{LSI} with C_{PI} .

2.3 Diffusion Processes

One sub-family of Markov processes is identified by the so-called diffusion processes, which represent an important modeling tool in several fields. Some of the theory is treated in [24], [23], [16] and [22].

2.3.1 Stochastic Differential Equations

Diffusion processes are solutions to stochastic differential equations, where the random component is typically modeled by the standard Brownian Motion (BM) or, more generally, by stochastic integrals with respect to the Brownian Motion. As a matter of fact, Brownian Motion is the first and most fundamental diffusion process ever studied. In 1817, the Scottish botanist Robert Brown observed under a microscope the “rapid oscillatory motion” of microscopic particles within pollen grains suspended in water. The mathematical formulation of this phenomenon was first provided by A. Einstein and later refined by N. Wiener, establishing Brownian Motion as the most famous stochastic process characterized by independent and normally distributed increments, which are also stationary. The Brownian Motion $(B_t)_{t \geq 0}$ in $\mathbb{R}^d$ consists of d independent standard real Brownian Motions. Assuming $B_0 = 0$ then $B_t \sim \mathcal{N}(0, tI_d)$. Its infinitesimal generator is $\mathcal{L} = \frac{1}{2}\Delta$, but usually we consider the time rescaling version $(\tilde{B}_t)_{t \geq 0} = (B_t)_{t \geq 0}$ so that the generator is the laplacian. The invariant and reversible measure of such a process is the Lebesgue one (dx) given that as $t \rightarrow \infty$ the Gaussian distribution flattens out, so that the dynamics visits the whole state-space; this makes the Brownian Motion not ergodic.

The associated Markov (Brownian) semigroup $(P_t)_{t \geq 0}$ admits kernel densities with respect to the Lebesgue, $p_t(x, y) = (4\pi t)^{-d/2} \exp(-|x - y|^2/4t)$, $t > 0$, $(x, y) \in \mathbb{R}^d \times \mathbb{R}^d$. The kernels satisfy the parabolic heat equation $\partial_t p_t = \Delta p_t$; this is why the Brownian semigroup is also called Heat semigroup.

Informally, the term *diffusion* is associated to Markov processes having continuous trajectories and that can be characterized through their infinitesimal generators. As a matter of fact, there are two ways of studying diffusion processes. The first approach is the probabilistic one, where we work with a SDE and consider each of its possible sample paths; the second approach is the analytical one, where we consider the Fokker-Planck PDE and its description of the density evolution. Actually, the two methods are not mutually independent and are linked by the Markov semigroup and its infinitesimal generator.

The general formulation of a stochastic differential equation is given by,

$$dX_t = b(X_t)dt + \sigma(X_t)dB_t, \quad X_0 = x, \quad (2.17)$$

where $X_t, B_t, b(\cdot) \in \mathbb{R}^d$ and $\sigma(\cdot) \in \mathbb{R}^{d \times d}$. As standard practice, we can not divide by dt due to the non-differentiability of the Brownian Motion. The most proper notation for such a dynamics is given by the stochastic integral equation,

$$X_t = X_0 + \int_0^t b(X_s)ds + \int_0^t \sigma(X_s)dB_s, \quad t \geq 0. \quad (2.18)$$

An important issue is related with existence and uniqueness of a stochastic process solving such equations. For what concerns existence we must require a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ on which we define a d -dimensional Brownian Motion $(B_t)_{t \geq 0}$. On the other hand, uniqueness might be a more delicate issue. We say that $(X_t^x)_{t \geq 0}$ is a solution to (2.17) if it satisfies the equation path-wise, namely for almost every $\omega \in \Omega$. Then uniqueness implies that, for any initial value x , two solutions X and X' of (2.17) are such that the maps $t \mapsto X_t$ and $t \mapsto X'_t$ are equal almost everywhere. The following theorem formalizes the existence and uniqueness of solutions to (2.17).

Theorem 5. *Assume that the maps $\mathbb{R}^d \ni x \mapsto \sigma(x)$ and $\mathbb{R}^d \ni x \mapsto b(x)$ have a linear growth,*

$$\|\sigma(x)\| + \|b(x)\| \leq D(1 + \|x\|), \quad (2.19)$$

for some finite constant D and all $x \in \mathbb{R}^d$ and are locally Lipschitz, meaning that for any compact set $K \subset \mathbb{R}^d$, there exists a constant $C_K > 0$ such that,

$$\|\sigma(x) - \sigma(y)\| + \|b(x) - b(y)\| \leq C_K \|x - y\|, \quad (2.20)$$

for all $x, y \in K$. Then, there exists a unique solution $(X_t^x)_{t \geq 0}$ to (2.17) defined on the time interval $[0, \infty)$.

Then the unique solution to (2.17), i.e. $(X_t^x)_{t \geq 0}$, will be a Markov process with continuous trajectories, adapted to the filtration $(\mathcal{F}_t)_{t \geq 0}$ and will satisfy certain properties such as the strong Markov property. Moreover it is associated with its own Markov semigroup $(P_t)_{t \geq 0}$ defined by $P_t f(x) = \mathbb{E}[f(X_t^x)]$. Furthermore, the infinitesimal generator is given by the second order operator,

$$\mathcal{L}f = \frac{1}{2} \sum_{i,j=1}^d (\sigma \sigma^T)_{ij} \partial_{ij}^2 f + \sum_{i=1}^d b_i \partial_i f. \quad (2.21)$$

A natural question, then, is to ask whether the process $(f(X_t^x))_{t \geq 0}$ satisfies a SDE as well. The answer is yes and it is given by the famous Itô's formula.

Theorem 6. *Let us consider the process $(f(X_t^x))_{t \geq 0}$, where $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is a smooth function, at least belonging to $\mathcal{C}^2(\mathbb{R}^d; \mathbb{R})$, and $(X_t^x)_{t \geq 0}$ is a solution to (2.17). Then, we have,*

$$df(X_t) = \sum_{i,j=1}^d \sigma_i^j(X_t) \partial_i f(X_t) dB_t^j + \mathcal{L}f(X_t) dt, \quad (2.22)$$

$$= dM_t^f + \mathcal{L}f(X_t) dt, \quad X_0 = x, \quad (2.23)$$

where the operator $\mathcal{L}$ is as in (2.21) and $(M_t^f)_{t \geq 0}$ is a local martingale.

It is then straightforward to understand that the generator models the infinitesimal deterministic evolution of the expected value of a smooth function f applied to the process, given the present state. The stochastic integral equation corresponding to (2.23) reads,

$$f(X_t^x) = M_t^f + \int_0^t \mathcal{L}f(X_s^x) ds, \quad t \geq 0. \quad (2.24)$$

This shows that diffusion processes solving SDEs are always *semi-martingales*. Assum-

ing the local martingale to be a true martingale, we have, for every $t \geq 0$,

$$\mathbb{E}[f(X_t^x)] = f(x) + \frac{1}{2} \int_0^t \mathbb{E}[\mathcal{L}f(X_s^x)] ds, \quad (2.25)$$

where the first term is given by the martingale equality condition, i.e. $\mathbb{E}[M_t^f | \mathcal{F}_s] = M_s^f$. Equation (2.25) tells us that the process $(X_t^x)_{t \geq 0}$ is a Diffusion process with generator $\frac{1}{2}\mathcal{L}$. In fact, setting $P_t f(x) = \mathbb{E}[f(X_t^x)] = \mathbb{E}[f(X_t) | X_0 = x]$, $\forall t \geq 0$, it holds that,

$$P_t f(x) = f(x) + \frac{1}{2} \int_0^t P_s \mathcal{L}f(x) ds, \quad t \geq 0. \quad (2.26)$$

This coincides with equation (2.2), derived previously. And as we already highlighted, $P_t f$ is the solution of the heat equation $\partial_t P_t f = \frac{1}{2} \mathcal{L} P_t f$ starting from f .

2.3.2 Langevin Diffusion

A key role in sampling theory is played by the so-called *Overdamped Langevin diffusion*, which is a general diffusion process used in physics, for example, to model the dynamics of particles undergoing diffusion due to both deterministic (here identified by the potential field) and stochastic (a simple stochastic integral with respect to the Brownian Motion) forces. The Langevin diffusion, together with its associated Fokker-Planck equation, describe a physical system from both a microscopic and a macroscopic point of view.

Definition 11. A stochastic process $(X_t)_{t \geq 0}$ is a Langevin diffusion with potential field $V \in \mathcal{C}^2(\mathbb{R}^d; \mathbb{R})$, if it solves the following stochastic differential equation (SDE),

$$dX_t = -\nabla V(X_t) dt + \sqrt{2} dB_t, \quad X_0 \sim \pi_0, \quad (2.27)$$

where $(B_t)_{t \geq 0}$ is a standard d -dimensional Brownian Motion.

Does the Langevin SDE have a unique strong solution? Well, as a result of theorem (5), if ∇V is Lipschitz, that is V is smooth, then there is a unique strong solution, namely, given a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F})_{t \geq 0}, \mathbb{P})$ and a d -dimensional standard Brownian Motion defined on it then there is a unique adapted process $(X_t)_{t \geq 0}$ having a modification with continuous trajectories and satisfying the integral equation $X_t = X_0 - \int_0^t \nabla V(X_s) ds + \sqrt{2} B_t$, $\forall t \geq 0$.

The drift term of the Langevin SDE is a pure gradient flow,

$$\dot{X}_t = -\nabla V(X_t),$$

which is an ODE targeting a minimizer of f . Consequently, Langevin diffusion can be interpreted as "a gradient flow + noise". However, such a dynamics converges in law to the stationary distribution of the process, $\pi \propto e^{-V(x)}$, which is unique and associated to the following *Fokker-Planck* equation,

$$\frac{\partial}{\partial t} \pi_t = \Delta \pi_t + \nabla \cdot (\pi_t \nabla V(x)), \quad X_0 \sim \pi_0, \quad (2.28)$$

$$= \mathcal{L}^* \pi_t. \quad (2.29)$$

Notice that the infinitesimal generator for the Langevin diffusion is $\mathcal{L} = \Delta - \nabla V \cdot \nabla$ and $\mathcal{L} = \mathcal{L}^*$ only on the space $\mathbb{L}^2(\pi)$. Note that $-\nabla V = \nabla \log \pi$, so that the process uses information about the the shape/geometry of the distribution to move toward higher density regions of π . Then, the noise term is essential to prevent the process from getting stuck in modes and to ensure proper exploration of the entire support of π .

As we said, Langevin diffusion is a general family of stochastic processes; if, for example, we set $V = \frac{1}{2} \|\cdot\|^2$ then we would have the *Ornstein–Uhlenbeck (OU)* process having as stationary measure the d-dimensional standard Gaussian, γ^d .

Going back to the optimization-sampling connection, [15] showed that if we consider the metric space $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$, where $\mathcal{P}_2(\mathbb{R}^d)$ is the space of probability measures with finite second moment defined over the d-dimensional Euclidean space and $\mathcal{W}_2$ is the 2-Wasserstein distance, then the Fokker-Planck dynamics, namely, the curve of measures $t \mapsto \pi_t$, can be interpreted as the gradient flow of the *Kullback–Leibler divergence*, $\text{KL}(\cdot \parallel \pi)$, with respect to 2-Wasserstein distance. Therefore, sampling consists in an optimization problem in a suitable space of measures; in fact, Langevin dynamics is the steepest descent flow that transports any initial distribution of the system to the invariant measure defined by the potential field.

Previously, we introduced two functional inequalities, namely the Poincaré inequality and the log-Sobolev one, in order to characterize properties satisfied by the stationary measure which guarantee an exponential convergence of the curve $t \mapsto \pi_t$ to π . For a

Langevin diffusion $(X_t)_{t \geq 0}$ defined on $\mathbb{R}^d$ the PI reads,

$$\text{Var}_\pi f \leq C_{PI} \mathbb{E}_\pi [\|\nabla f\|^2], \quad (2.30)$$

for all functions $f : \mathbb{R}^d \rightarrow \mathbb{R}$. For $E = \mathbb{R}^d$ it is even more explicit the fact that the Dirichlet form measures the smoothness of a function. Instead, the LSI reads,

$$\text{Ent}_\pi(f^2) \leq 2C_{LSI} \mathbb{E}_\pi [\|\nabla f\|^2], \quad \text{or equivalently} \quad (2.31)$$

$$\text{KL}(\pi_t \| \pi) \leq 2C_{LSI} \mathbb{E}_\pi [\|\nabla \sqrt{\rho_t}\|^2], \quad (2.32)$$

where we recall that $\rho_t = d\pi_t/d\pi$ is the Radon-Nikodym derivative of $\pi_t = \pi_0 P_t$ with respect to π . Notice that $\mathbf{P}$ is self-adjoint due to the reversibility property of the Langevin diffusion with respect to its stationary distribution.

The class of measures satisfying LSI and/or PI is quite large, including all strongly log-concave measures (due the Bakry–Emery criterion) and, for the PI, all log-concave measures. In particular, LSI and Poincaré inequality are preserved under bounded perturbation and Lipschitz mapping whereas logconcavity would be destroyed. Given these properties, it is easy to exhibit examples of non-logconcave distributions satisfying LSI or Poincaré inequality; one possible example is obtained by taking a convex body and apply small perturbations to it so that it is not convex anymore but still satisfies isoperimetry. Take a look at Figure (1), taken from [30].

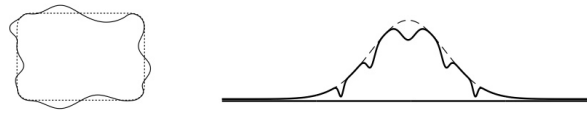


Figure 1: Illustrations of non-logconcave distributions satisfying LSI or Poincaré inequality: the uniform distribution on a nonconvex set (left), and a small perturbation of a logconcave distribution, e.g., Gaussian (right)).

Consider one last observation. Processes in continuous-time will be discretized in order to be applicable, as we will see in the next chapter. The way in which the continuous-time result affects its discretization counterpart operates through two channels. In fact, if the potential V is α -strongly convex then, if we consider two different Langevin diffusions, so that the corresponding Fokker-Planck describe the evolution of the maps $t \mapsto \nu_t$ and $t \mapsto \pi_t$ with starting point ν_0 and π_0 , then we have a contraction

of the $\mathcal{W}_2$ distance between the two processes, $\mathcal{W}_2^2(\nu_t, \pi_t) \leq e^{-2\alpha t} \mathcal{W}_2^2(\nu_0, \pi_0)$. At the same time, if the equilibrium distribution, π , of the Langevin diffusion satisfies a LSI with constant $C_{LSI} = 1/\alpha$, then along the dynamics $t \mapsto \pi_t$ we have the exponential decrease in relative entropy, $\text{KL}(\pi_t \| \pi) \leq e^{-2\alpha t} \text{KL}(\pi_0 \| \pi)$. Finally, if π satisfies the PI then the previous result holds for the chi-square divergence. Then, these results affect the discretization analysis because at initialization we typically have, for the first two cases, $\mathcal{W}_2^2(\pi_0, \pi), \text{KL}(\pi_0 \| \pi) = O(d)$ so that the Langevin dynamics reaches a ϵ -error in $O(\log(d/\epsilon))$ time. For the PI, instead, we have $\chi^2(\pi_0 \| \pi) = \exp(O(d))$ so that it takes $O(d \vee \log(1/\epsilon))$ time to reach the same amount of error. Secondly, if the result in continuous-time is given using a specific metric/divergence then it is likely to have the same metric/divergence for the discretization analysis.

3 Log-Concave Sampling

Sampling from a prescribed probability distribution is a fundamental task in probabilistic modelling. For example, samples allow us to approximate expectations by Monte Carlo methods. In Bayesian inference, the posterior density is

$$p(\theta \mid X_{1:n}) = \frac{p(\theta)p(X_{1:n} \mid \theta)}{p(X_{1:n})}, \quad p(X_{1:n}) = \int_{\Theta} p(\theta)p(X_{1:n} \mid \theta) \, d\theta.$$

When $\Theta \subseteq \mathbb{R}^d$ has large dimension, evaluating the normalising integral can be difficult. Sampling methods that use an unnormalised density are therefore useful in statistics, machine learning, physics, and finance. In practice, an algorithm often produces samples from an approximation to the target. Its accuracy must then be assessed using a specified distance or divergence, such as total variation, relative entropy, or the quadratic Wasserstein distance.

In this chapter, the target has density

$$\pi(x) = Z^{-1}e^{-V(x)}, \quad Z = \int_{\mathbb{R}^d} e^{-V(x)} \, dx < \infty.$$

We use π for both this density and its probability measure when the meaning is clear. We assume access to a first-order oracle that returns $V(x) - V(0)$ and $\nabla V(x)$. The unknown additive constant in V does not affect the algorithms. The next chapter considers a different information model: samples from the target are available for learning the time-dependent scores of a forward diffusion, while its density need not be available analytically. These are distinct assumptions about access to the target, rather than interchangeable solutions to the same computational problem.

The Wasserstein gradient-flow interpretation of Langevin dynamics connects sampling with optimisation. We explore this connection through Langevin Monte Carlo and then consider a Metropolis–Hastings correction. Unless stated otherwise, initial laws belong to $\mathcal{P}_2(\mathbb{R}^d)$, and every Gaussian innovation is independent of the past. The notation $a \lesssim b$ denotes an inequality up to a universal positive constant, and $a \asymp b$ denotes two such inequalities. Complexity bounds count iterations, not elapsed diffusion time; one ULA iteration requires one gradient evaluation.

3.1 Langevin Monte Carlo

Under the well-posedness and invariance assumptions discussed in the previous chapter, the Langevin diffusion

$$dX_t = -\nabla V(X_t) dt + \sqrt{2} dB_t$$

has invariant law π . A Poincaré inequality gives exponential decay in chi-squared divergence for initial densities with finite chi-squared divergence; a log-Sobolev inequality gives the corresponding relative-entropy bound for finite initial entropy. These continuous-time statements do not, by themselves, control numerical discretisation error.

The Euler–Maruyama discretisation is called *Langevin Monte Carlo* (LMC), or the *unadjusted Langevin algorithm* (ULA). The equation reads,

$$X_{(k+1)h} := X_{kh} - h\nabla V(X_{kh}) + \sqrt{2}(B_{(k+1)h} - B_{kh}), \quad (3.1)$$

assuming constant step-size $h > 0$. Naturally, as $h \downarrow 0$ we obtain the corresponding continuous-time dynamics. Let p_{kh} be the probability distribution of the k -th step of the ULA, that is $X_{kh} \sim p_{kh}$. It turns out that the stationary distribution of the ULA, π_h , is biased, that is, $\pi_h \neq \pi$. Consequently, differently from what happens for the Langevin diffusion, the relative entropy $KL(p_k \parallel \pi)$ does not tend to 0 along the ULA trajectory because of the bias term $KL(\pi_h \parallel \pi) > 0$. Therefore, when implementing such an algorithm we are approximately sampling from π ; here is why we would like to obtain some non-asymptotic bounds for the distance (usually the 2-Wasserstein one) between the algorithm distribution and the target one. However, non-asymptotic bounds can differ substantially from one setting to another as geometric properties of the target distribution highly influence the mixing capabilities of the ULA.

The algorithm works as follows.

1. Initialize $X_0 \sim p_0$; choose N and set $k = 0$

2. For $k = 0, \dots, N - 1$:

sample $Z_{(k+1)h} \sim \mathcal{N}(0, 2hI_d)$

set $X_{(k+1)h} \leftarrow X_{kh} - h\nabla V(X_{kh}) + Z_{(k+1)h}$

3. Output $(X_h, \dots, X_{Nh})$.

Such a trajectory consists of dependent samples. For suitable potentials and sufficiently small h , ULA has a unique invariant distribution π_h , but generally $\pi_h \neq \pi$. This is *discretisation bias*, not a failure of ergodicity. Under the conditions of the Markov-chain ergodic theorem, trajectory averages converge to expectations under π_h . They can therefore estimate target expectations approximately, with an error that includes stationary bias as well as finite-run error. Existence of π_h is not automatic for arbitrary potentials or step sizes.

A Gaussian example makes the distinction explicit. For $V(x) = \|x\|^2/2$, ULA becomes

$$X_{(k+1)h} = (1 - h)X_{kh} + \sqrt{2}(B_{(k+1)h} - B_{kh}).$$

For $0 < h < 2$, this chain is ergodic with invariant law

$$\pi_h = \mathcal{N}(0, (1 - h/2)^{-1}I_d),$$

whereas the target is $\mathcal{N}(0, I_d)$. Increasing the run length removes the initialisation error but does not remove this covariance discrepancy.

3.1.1 Wasserstein Coupling

One of the very first papers to introduce the similarities between gradient descent and the ULA was [10], where the author provides some theoretical convergence guarantees for the Langevin Monte Carlo method in the usual $\mathcal{W}_2$ metric, assuming smooth and strongly log-concave distributions. A nice analysis of ULA, given the preceding assumptions, is based on coupling together the continuous-time process with the corresponding discretization. Then, the coupling is used to measure the approximation error accumulated along the trajectories.

Theorem 7. For $k \in \mathbb{N}$, let p_{kh} be the probability distribution of the k -th iteration of the ULA with step size $h > 0$. Assume that the target $\pi \propto e^{-V}$ is α -strongly log-concave and β -smooth, i.e. $\alpha I_d \preceq \nabla^2 V \preceq \beta I_d$. Then, assuming $h \lesssim \frac{1}{\beta\kappa}$, for all $N \in \mathbb{N}$,

$$\mathcal{W}_2(p_{Nh}, \pi) \leq e^{-\frac{\alpha Nh}{2}} \mathcal{W}_2(p_0, \pi) + O\left(\frac{\beta\sqrt{dh}}{\alpha}\right). \quad (3.2)$$

If we set $h \asymp \frac{\epsilon^2}{\beta\kappa d}$, then for any $\epsilon \in (0, \sqrt{d}]$ we obtain $\sqrt{\alpha}\mathcal{W}_2(p_{Nh}, \pi) \leq \epsilon$ after

$$N = O\left(\frac{\kappa^2 d}{\epsilon^2} \log \frac{\sqrt{\alpha}\mathcal{W}_2(p_0, \pi)}{\epsilon}\right) \text{ iterations.}$$

The quantity $\kappa := \beta/\alpha$ is the condition number. The notation $h \asymp \epsilon^2/(\beta\kappa d)$ means that h is bounded above and below by positive constant multiples of this expression; for the theorem, the step-size constant must be chosen sufficiently small.

Equation (3.2) separates the error into two contributions: an initialisation term that decreases exponentially with the simulated time Nh , and a term controlling the accumulated discretisation error. Under the stated assumptions and step-size restriction, ULA converges to its own stationary distribution π_h . Taking the long-time limit therefore gives an upper bound on the stationary bias $\mathcal{W}_2(\pi_h, \pi)$.

A larger step size generally leads to a less accurate approximation of the continuous-time dynamics because the drift integral is replaced by a single gradient evaluation at each step. In this bound, the discretisation term grows as $\sqrt{dh}$, with the remaining parameters fixed. This is an upper bound: the actual bias can be smaller for particular targets.

The condition number also affects the convergence guarantee. A large κ indicates a substantial difference between the largest and smallest curvature bounds of the potential. This can reflect different curvatures across directions or across regions of the state space. The step size must be small enough to accommodate the largest curvature, while the smallest curvature controls the guaranteed contraction rate.

If the minimiser x_* of V , equivalently the mode of π , is available, it provides a convenient initial state. Under the theorem's assumptions, choosing $p_0 = \delta_{x_*}$ gives

$$\sqrt{\alpha}\mathcal{W}_2(p_0, \pi) \leq \sqrt{d}.$$

Finding or approximating this minimiser requires a separate optimisation step. Finally, the scaled distance $\sqrt{\alpha} \mathcal{W}_2$ is useful because it is invariant under a scalar change of spatial units, with the corresponding rescaling of α . We now give the proof of (7), as in [25].

Proof of Theorem 7. We assume that both the Langevin diffusion and the ULA are initialized at the same measure p_0 . We couple the two processes by taking $X_0 = Z_0$ and using the same Brownian Motion,

$$\begin{aligned} X_h &= Z_0 - h \nabla V(Z_0) + \sqrt{2} B_h, \\ Z_h &= Z_0 - \int_0^h \nabla V(Z_t) dt + \sqrt{2} B_h. \end{aligned}$$

Coupling means to introduce some kind of dependence structure (same starting point and same Brownian Motion here) between the two processes while maintaining the original marginal distributions. Then we have,

$$\begin{aligned} \mathcal{W}^2(p_h, \pi_h) &\leq \mathbb{E}[\|X_h - Z_h\|^2] \leq \mathbb{E} \left[\left\| \int_0^h \nabla V(Z_t) dt - h \nabla V(Z_0) \right\|^2 \right] \\ &\leq h \int_0^h \mathbb{E}[\|\nabla V(Z_t) - \nabla V(Z_0)\|^2] dt. \end{aligned}$$

Now, given that V is β -smooth, i.e. $\|\nabla V(Z_t) - \nabla V(Z_0)\|^2 \leq \beta^2 \|Z_t - Z_0\|^2$, we need to bound the movement of the Langevin diffusion in time t , namely

$$\|Z_t - Z_0\| = - \int_0^t \nabla V(Z_s) ds + \sqrt{2} B_t.$$

As a result of stochastic calculus theory, if ∇V is β -Lipschitz and $t \leq 1/3\beta$ then,

$$\mathbb{E}[\|Z_t - Z_0\|^2] \leq 8t^2 \mathbb{E}[\|\nabla V(Z_0)\|^2] + 8dt$$

and,

$$\mathcal{W}_2^2(p_h, \pi_h) \leq \beta^2 h \int_0^h \mathbb{E}[\|Z_t - Z_0\|^2] dt \leq 3\beta^2 h^4 \mathbb{E}[\|\nabla V(Z_0)\|^2] + 4\beta^2 dh^3.$$

Therefore, we managed to bound the discretization error for one step of the ULA.

Now we produce a coupling of $p_{(k+1)h}$ and π . Assume $X_{kh} \sim p_{kh}$ and $Z_{kh} \sim \pi$ to be optimally coupled, that is such that they minimize their $\mathcal{W}_2$ -distance. Using the same Brownian Motion we set,

$$\begin{aligned} X_{(k+1)h} &:= X_{kh} - h\nabla V(X_{kh}) + \sqrt{2}(B_{(k+1)h} - B_{kh}), \\ Z_t &:= Z_{kh} - \int_{kh}^t \nabla V(Z_s) \mathbf{d}s + \sqrt{2}(B_t - B_{kh}), \quad \text{for } t \in [kh, (k+1)h]. \end{aligned}$$

Then, $X_{(k+1)h} \sim p_{(k+1)h}$ and $Z_t \sim \pi$ for $t \geq kh$ due to stationarity. We define an auxiliary process,

$$\bar{X}_t := X_{kh} - \int_{kh}^t \nabla V(\bar{X}_s) \mathbf{d}s + \sqrt{2}(B_t - B_{kh}), \quad t \in [kh, (k+1)h],$$

which denotes the Langevin diffusion started at X_{kh} . Then, we bound,

$$\begin{aligned} \mathcal{W}_2(p_{(k+1)h}, \pi) &\leq \sqrt{\mathbb{E} [\|X_{(k+1)h} - Z_{(k+1)h}\|^2]} \\ &\leq \sqrt{\mathbb{E} [\|\bar{X}_{(k+1)h} - Z_{(k+1)h}\|^2]} + \sqrt{\mathbb{E} [\|X_{(k+1)h} - \bar{X}_{(k+1)h}\|^2]}. \end{aligned}$$

For the first term, which consists of two Langevin diffusion with an α -strongly convex potential, we have the following contraction inequality,

$$\begin{aligned} \mathbb{E} [\|\bar{X}_{(k+1)h} - Z_{(k+1)h}\|^2] &\leq e^{-2\alpha h} \mathbb{E} [\|X_{kh} - Z_{kh}\|^2] \\ &= e^{-2\alpha h} \mathcal{W}_2^2(p_{kh}, \pi). \end{aligned}$$

For the second term we are back to the setting we analyzed for the one step discretization bound. Thus,

$$\begin{aligned} \mathbb{E} [\|X_{(k+1)h} - \bar{X}_{(k+1)h}\|^2] &\leq 3\beta^2 h^4 \mathbb{E} [\|\nabla V(X_{kh})\|^2] + 4\beta^2 d h^3 \\ &\lesssim \beta^4 h^4 \mathbb{E} [\|X_{kh} - Z_{kh}\|^2] + \beta^2 h^4 \mathbb{E}_\pi [\|\nabla V\|^2] + \beta^2 d h^3 \\ &\leq \beta^4 h^4 \mathcal{W}_2^2(p_{kh}, \pi) + \beta^3 d h^4 + \beta^2 d h^3, \end{aligned}$$

where we used the fact that if V is β -smooth then $\mathbb{E}_\pi [\|\nabla V\|^2] \leq \beta d$. Dropping the second term given the assumption $h \lesssim 1/\beta\kappa$ and using the relation $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$

for $x, y \geq 0$, we obtain,

$$\mathcal{W}_2(p_{(k+1)h}, \pi) \leq e^{-\alpha h} \mathcal{W}_2(p_{kh}, \pi) + O(\beta^2 h^2 \mathcal{W}_2(p_{kh}, \pi) + \beta d^{1/2} h^{3/2}).$$

Provided that $h \lesssim 1/\beta\kappa$ we have $e^{-\alpha h} + O(\beta^2 h^2) \leq e^{-\alpha h/2}$, so that,

$$\mathcal{W}_2(p_{(k+1)h}, \pi) \leq e^{-\alpha h/2} \mathcal{W}_2(p_{kh}, \pi) + O(\beta d^{1/2} h^{3/2}).$$

Then, we can iterate this construction to obtain,

$$\mathcal{W}_2(p_{Nh}, \pi) \leq e^{-\alpha Nh/2} \mathcal{W}_2(p_0, \pi) + O\left(\frac{\beta d^{1/2} h^{1/2}}{\alpha}\right),$$

completing the proof. $\square$

This is a nice and intuitive proof, which relies on strong log-concavity of the stationary measure given the use of the contraction property of the Langevin gradient flow in the Wasserstein metric. However, this proof does not hold under weaker assumptions such as LSI and, at the same time, it does not provide sharp, tight non-asymptotic bounds. The next proof gives tight dependence of LMC on the condition number κ of the equilibrium distribution π .

3.1.2 Convex Optimization

This second proof is the result of considering the ULA as a first-order optimization algorithm in the space of measures and use tools from convex optimization. A first treatment of this approach is given in [32]; the finest results are then obtained in [12]. Here we will assume $\pi = \exp(-V)$ thus V incorporates the normalizing constant as well. The starting point is to decompose the relative divergence in two components,

$$\text{KL}(\mu \parallel \pi) := \int_{\mathbb{R}^d} V d\mu + \int_{\mathbb{R}^d} \mu \ln \mu,$$

where the first component, $\mathbb{E}_\mu[V]$, is usually called the *energy functional* and the second one is the usual negative entropy functional. Then, π satisfies a variational principle, namely it minimizes, over all probability measures on $E = \mathbb{R}^d$, the above functional, $\text{KL}(\cdot \parallel \pi)$, which is also called *free energy functional*. This is a consequence

of what we already pointed out in the previous chapters, namely that Langevin diffusion is the gradient flow of the free energy functional in the metric space $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$. Correspondingly, we divide the ULA dynamics in two parts,

$$\begin{aligned} X_{kh}^+ &:= X_{kh} - h\nabla V(X_{kh}), \\ X_{(k+1)h} &:= X_{kh}^+ + \sqrt{2}(B_{(k+1)h} - B_{kh}). \end{aligned}$$

The first step is a simple gradient descent on the potential V . It can be shown that $p_{kh}^+ := \text{law}(X_{kh}^+)$ is obtained, in the space of measures, from p_{kh} by taking a gradient step of the energy functional $\mathcal{E}(\mu) := \int_{\mathbb{R}^d} V d\mu$ with respect to the geometry induced by the 2-Wasserstein distance. The second step is the heat flow (the name comes from the fact the Fokker-Planck of the Brownian Motion is the heat equation) that is a Wasserstein gradient flow for the entropy functional $\mathcal{H}(\mu) := \int_{\mathbb{R}^d} \mu \ln \mu$, in the space of measures. This way of dividing the dynamics into its components (linearly here) and then discretize both of them is known as a splitting scheme. The LMC algorithm is also called the "forward-flow" algorithm due to the fact that the gradient descent step is called the forward method in optimization while the second step is the (heat)-flow. Then, this scheme aims to obtain sharp bounds by showing that the forward step dissolves the energy, $\mathcal{E}$, without increasing substantially the entropy, $\mathcal{H}$ and that the flow step does the same but in the other direction. Let's state the results.

Lemma 3. *Let $\pi = \exp(-V)$ be the target distribution satisfying $0 \preceq \alpha I_d \preceq \nabla^2 V \preceq \beta I_d$. Let $(p_{kh})_{k \in \mathbb{N}}$ denote the probability measures defined by the ULA with step-size $h \in [0, \beta^{-1}]$. Then,*

$$\text{KL}(p_{(k+1)h} \parallel \pi) \leq \frac{1 - \alpha h}{2h} \mathcal{W}_2^2(p_{kh}, \pi) - \mathcal{W}_2^2(p_{(k+1)h}, \pi) + 2\beta d h^2. \quad (3.3)$$

From this lemma we obtain the following theorem, due to [12].

Theorem 8. *Let $\pi = \exp(-V)$ be the target distribution satisfying $0 \preceq \alpha I_d \preceq \nabla^2 V \preceq \beta I_d$. Let $(p_{kh})_{k \in \mathbb{N}}$ denote the probability measures defined by the ULA. Then:*

1. *(V convex, $\alpha = 0$) For any $\epsilon \in [0, \sqrt{d}]$, if we set $h \asymp \epsilon^2/\beta d$, then for the mixture*

distribution $\bar{p}_{Nh} := N^{-1} \sum_{k=1}^N p_{kh}$ it holds that,

$$\sqrt{\text{KL}(\bar{p}_{Nh} \parallel \pi)} \leq \epsilon, \text{ with mixing time} \quad (3.4)$$

$$t_{\text{mix}}(\epsilon, p_0) = O\left(\frac{\beta d \mathcal{W}_2^2(p_0, \pi)}{\epsilon^4}\right). \quad (3.5)$$

2. (V strongly-convex, $\alpha > 0$) Then, for any $\epsilon \in [0, \sqrt{d}]$, if we set $h \asymp \epsilon^2/\beta d$, we obtain

$\sqrt{\alpha} \mathcal{W}_2(p_{Nh}, \pi) \leq \epsilon$ and $\sqrt{\text{KL}(\bar{p}_{Nh, 2Nh} \parallel \pi)} \leq \epsilon$ with mixing time

$$t_{\text{mix}}(\epsilon, p_0) = O\left(\frac{\kappa d}{\epsilon^2} \log\left(\frac{\sqrt{\alpha} \mathcal{W}_2(p_0, \pi)}{\epsilon}\right)\right), \quad (3.6)$$

where $\bar{p}_{Nh, 2Nh} := N^{-1} \sum_{k=N+1}^{2N} p_{kh}$.

Thus, considering the strongly log-concave case, we managed to reduce the computational cost, that is the mixing time, by a factor equal to the condition number κ , with respect to the proof we gave with the 2-wasserstein coupling.

The LMC algorithm can be applied in several contexts having different assumptions. For example, joining the work of [30] and [8] we can obtain, under the assumptions that π satisfies a LSI and V is β -smooth, sampling guarantees in relative entropy that scale polynomially with respect to the parameters, e.g. $N = O\left(\frac{C_{LSI}^2 \beta^2 d}{\epsilon^2} \log \frac{\text{KL}(p_0 \parallel \pi)}{\epsilon^2}\right)$. This setting covers cases where π is not log-concave as long as the above assumptions are met.

3.2 Metropolis-Hastings

The Langevin Monte Carlo algorithm is the most basic sampling algorithm given that it only consists in a discretization of the corresponding continuous-time dynamics. Furthermore, given the presence of the bias introduced by the discretization, the algorithm scales as $\text{poly}(1/\epsilon)$ meaning that the more precise we want to be in approximating π the higher, as a polynomial function of $1/\epsilon$, the computational cost (that is the number of iterations needed by the algorithm) we have. This is why these kinds of samplers are called *low-accuracy samplers*.

In order to obtain algorithms that scale as $\text{polylog}(1/\epsilon)$, i.e. they scale polynomially

in the logarithm of the argument, we need to remove the bias; this is achieved by *high-accuracy samplers*, e.g. by applying the Metropolis–Hastings filter to the ULA and obtain the *Metropolis-adjusted Langevin algorithm (MALA)*, which was introduced by [3]. The class of Metropolis-Hastings (MH) algorithms is an established family of Markov Chain Monte Carlo (MCMC) algorithms used to sample from a target distribution π by constructing a π -reversible Markov Chain. A first formulation was developed in 1953 by the physicist Nicholas Metropolis and then generalized by the statistician W.K. Hastings. An instance of such computational models is given by the Gibbs sampler, which is suitable for sampling from a multivariate distribution by iteratively sampling from its univariate full conditional distributions. These algorithms apply an idea similar to Rejection Sampling, that is given the current state of the chain they propose a potential new sample in the chain from a proposal kernel Q and then either accept or reject the new move with a well-defined probability which ensures the chain to be π -stationary. Let Q be a kernel on the product space $\mathbb{R}^d \times \mathbb{R}^d$ so that $Q(x, \cdot)$ is a probability measure on the d -dimensional Euclidean space, for all $x \in \mathbb{R}^d$. Usually, we choose kernels such that $Q(x, \cdot)$ is absolutely continuous with respect to the Lebesgue measure and will write $Q(x, y)$ to indicate the corresponding density evaluated at $y \in \mathbb{R}^d$. Thus, we start from $X \in \mathbb{R}^d$ and propose a new sample $Y \sim Q(X, \cdot)$; then we accept Y with an acceptance probability equal to $a(X, Y)$. Usually, the choice for such an acceptance probability is given by the Metropolis-Hastings filter,

$$a(X, Y) := \min \left\{ 1, \frac{\pi(y)Q(y, x)}{\pi(x)Q(x, y)} \right\}. \quad (3.7)$$

Then, the transition kernel $P : \mathbb{R}^d \times \mathcal{B}(\mathbb{R}^d) \rightarrow [0, 1]$ of the resulting Markov Chain $(X_n)_{n \geq 0}$ is given by,

$$P(x, dy) = Q(x, dy)A(x, y) + \left(1 - \int_{\mathbb{R}^d} Q(x, dy')A(x, y') \right) \delta_x(dy), \quad (3.8)$$

where the integral computes the total probability of accepting a new move so that inside the parenthesis we have the rejection probability and $\delta_x(dy)$ is the Dirac measure. Then, the Chapman-Kolmogorov equations allow us to compute the k -th step transition kernel as $P^k(x, dy) = \int_{z \in \mathbb{R}^d} P^{k-1}(x, dz)P(z, dy)$. It turns out that the Markov chain we just defined through its transition kernel P , which depends on the proposal kernel Q and on

the acceptance probability $a(\cdot, \cdot)$, is π -reversible, that is,

$$\forall x, y \in \mathbb{R}^d, x \neq y : \pi(x)P(x, y) = \pi(y)P(y, x),$$

which is a stronger condition than π -stationarity, which can be obtained by summing both sides,

$$\begin{aligned} \pi(x) &= \sum_{y \in \mathbb{R}^d} \pi(x)P(x, y) \\ &= \sum_{y \in \mathbb{R}^d} \pi(y)P(y, x). \end{aligned}$$

Furthermore, a π -reversible Markov chain is also time-reversible, that is the process defined as $Y_N = X_0, \dots, Y_0 = X_N$ is a Markov chain with the same transition probabilities. Finally, we can define the transition operator $\mathcal{T}_P$ as,

$$\mathcal{T}_P(p_n)(S) := \int_{y \in \mathbb{R}^d} \mathrm{d}p_n(y)P(y, A), \quad \forall A \in \mathcal{B}(\mathbb{R}^d),$$

so that if p_n is the distribution of the current state then $\mathcal{T}_P(p_n) = p_{n+1}$ and, recursively, $\mathcal{T}_{P^k}(p_n) = p_{n+k}$.

Therefore, the Metropolis-Hastings filter (consisting on the acceptance-rejection step) makes the sampling algorithm unbiased, no matter the choice of the proposal kernel. The proposal kernel will influence the velocity at which the algorithm converges to stationarity. One important thing is that here we are assuming the chain to be irreducible, recurrent and aperiodic, that is ergodic, so that the stationary distribution is unique and we can apply the ergodic theorem (after a burn-in period) which can be interpreted as a law of large numbers for Markov chains. Now, in order to implement the MH algorithm we need to be able to easily sample from $Q(x, \cdot)$ and to evaluate its density for the acceptance probability.

The choice of the proposal, as we said, allows for some flexibility. Among the options we can find: the Independent MH algorithm where the proposal is independent on the current value of the chain, the Metropolized random walk (MRW) where the proposal is a d-dimensional Gaussian centered at the current state with diagonal covariance matrix, Metropolized Hamiltonian Monte Carlo (MHMC) and the Metropolis-adjusted

Langevin algorithm (MALA) where the proposal is given by the LMC algorithm, namely $Q(x, \cdot) = \mathcal{N}(\cdot; x - h\nabla V(x), 2hI_d)$.

3.2.1 Metropolis-adjusted Langevin algorithm

The general assumptions are always that the potential field, V , of $\pi \propto e^{-V}$, is α -strongly convex and β -smooth, i.e. $\alpha I_d \preceq \nabla^2 V \preceq \beta I_d$.

Then, the procedure of the MALA is as follows. Choose the step-size $h > 0$ (this choice depends on several variables), the total number of iterations N and set $n = 0$. Initialize the algorithm at $X_0 \sim p_0$. Then, for $n = 0, \dots, N - 1$, repeat:

1. Proposal step: sample $Y_{n+1} \sim Q(X_n, \cdot)$, where

$$Q(X_n, \cdot) := (4\pi h)^{-d/2} \exp\left(-\frac{\|\cdot - X_n + h\nabla V(X_n)\|^2}{4h}\right),$$

2. Accept-reject step: with probability $\min\left\{1, \frac{\pi(Y_{n+1})Q(Y_{n+1}, X_n)}{\pi(X_n)Q(X_n, Y_{n+1})}\right\}$ set $X_{n+1} \leftarrow Y_{n+1}$, else set $X_{n+1} \leftarrow X_n$,
3. Output: $(X_1, \dots, X_N)$.

Now, a delicate role is played by the starting point, $X_0 \sim p_0$. There are two options: a feasible start and a warm one. Basically, the difference is given by how close the initial distribution is to the stationary one. Usually, we take $p_0 = \text{normal}(0, \beta^{-1}I_d)$ with the minimizer of V being 0, $\nabla V(0) = 0$. For the warm start, instead, we assume to have found p_0 such that $\chi^2(p_0 \parallel \pi) = O(1)$ (remember what we said at the end of chapter 2 about the usual initial complexity under a PI, $\chi^2(p_0 \parallel \pi) = \exp(O(d))$, so that this is indeed an improvement). We are using the chi-square divergence because Markov chains are usually analyzed via spectral theory which is related to the PI - Spectral Gap inequality.

Now, we can state the sampling guarantees for both the situations. For the feasible start setting you can take a look at [13].

Theorem 9. *Consider the target distribution $\pi \propto e^{-V}$ satisfying the assumptions previously stated. Then, given $X_0 \sim \mathcal{N}(0, \beta^{-1}I_d)$ and appropriate step-size $h > 0$, the MALA*

returns a measure p_N satisfying,

$$\sqrt{\chi^2(p_N \parallel \pi)} \leq \epsilon \quad \text{after} \quad (3.9)$$

$$N = \tilde{O} \left(\kappa d \operatorname{polylog} \frac{1}{\epsilon} \right) \quad \text{iterations.} \quad (3.10)$$

For the result with a warm start you can consult [9], [33] and [6].

Theorem 10. Consider the target distribution $\pi \propto e^{-V}$ satisfying the assumptions previously stated. Then, given an initial distribution p_0 satisfying $\chi^2(p_0 \parallel \pi) = O(1)$ and appropriate step-size $h > 0$, the MALA returns a measure p_N satisfying,

$$\|p_N - \pi\|_{TV} \leq \epsilon \quad \text{after} \quad (3.11)$$

$$N = \tilde{O} \left(\kappa d^{1/2} \operatorname{polylog} \frac{1}{\epsilon} \right) \quad \text{iterations.} \quad (3.12)$$

Therefore, then we have an improvement of magnitude $d^{1/2}$ in the scaling of the computational cost that, for high-dimensional distributions, can implicate remarkable differences.

3.3 More Computational Models

The sampling methods presented so far are part of a broader class of algorithms based on stochastic dynamics. One approach to improving their computational efficiency is to use a more accurate discretization scheme. For example, the randomized midpoint method replaces the Euler approximation of the drift integral with an evaluation at a randomly selected intermediate time.

For an exact Langevin trajectory, if $u_k \sim \text{Uniform}([0, 1])$ is independent of the process, then

$$\mathbb{E}_{u_k} \left[h \nabla V(X_{(k+u_k)h}) \mid (X_t)_{kh \leq t \leq (k+1)h} \right] = \int_{kh}^{(k+1)h} \nabla V(X_t) dt.$$

Thus, the random-time evaluation is an unbiased estimator of the drift integral, conditional on the exact trajectory. However, the intermediate state is not directly available and must itself be approximated. Using an Euler–Maruyama step gives the randomized

midpoint Langevin Monte Carlo (RM-LMC) scheme:

$$\begin{aligned} X_{(k+u_k)h}^+ &:= X_{kh} - u_k h \nabla V(X_{kh}) + \sqrt{2}(B_{(k+u_k)h} - B_{kh}), \\ X_{(k+1)h} &:= X_{kh} - h \nabla V(X_{(k+u_k)h}^+) + \sqrt{2}(B_{(k+1)h} - B_{kh}), \end{aligned}$$

where $(u_k)_{k \in \mathbb{N}}$ are independent and uniformly distributed on $[0, 1]$, independently of X_0 and the Brownian motion. The superscript $+$ distinguishes the intermediate approximation from the exact diffusion state. Replacing the exact state by this approximation does not, in general, preserve exact unbiasedness.

The two Brownian increments within each iteration overlap and must therefore be sampled jointly. An implementation uses independent Gaussian vectors $\xi_k, \eta_k \sim \mathcal{N}(0, I_d)$, independent also of u_k and the previous iterations, and sets

$$\begin{aligned} B_{(k+u_k)h} - B_{kh} &= \sqrt{u_k h} \xi_k, \\ B_{(k+1)h} - B_{kh} &= \sqrt{u_k h} \xi_k + \sqrt{(1 - u_k)h} \eta_k. \end{aligned}$$

Each RM-LMC iteration requires two gradient evaluations.

Assume, as before, that

$$0 < \alpha I_d \preceq \nabla^2 V(x) \preceq \beta I_d, \quad x \in \mathbb{R}^d,$$

and write $\kappa = \beta/\alpha$. With an appropriately chosen step size satisfying $h \lesssim 1/(\beta\sqrt{\kappa})$, the RM-LMC analysis gives $\sqrt{\alpha}\mathcal{W}_2(p_{Nh}, \pi) \leq \epsilon$ after

$$N = \tilde{O} \left(\frac{\kappa d^{1/2}}{\epsilon} \vee \frac{\kappa^{4/3} d^{1/3}}{\epsilon^{2/3}} \right), \quad 0 < \epsilon \leq \frac{\sqrt{d}}{\kappa^{1/4}},$$

where p_{Nh} denotes the law of the RM-LMC iterate X_{Nh} , with an initial law having finite second moment; see [25]. Here $\vee$ denotes the maximum, and $\tilde{O}$ suppresses logarithmic factors, including the dependence on the initial distance and the accuracy. This bound improves the dependence on dimension and accuracy in suitable regimes compared with Euler-based LMC. It does not imply an improvement in every parameter: one of the terms depends on $\kappa^{4/3}$.

Another approach changes the underlying dynamics. Hamiltonian Monte Carlo

introduces momentum variables and uses Hamilton’s equations, an equivalent formulation of classical mechanics, to construct proposals. Practical implementations typically combine numerical integration with momentum refreshment and a Metropolis–Hastings correction. Underdamped Langevin diffusion, also called kinetic Langevin diffusion, introduces friction and stochastic forcing into the momentum dynamics. It is described by coupled equations for position and momentum, with the target distribution as the position marginal of its invariant law under suitable assumptions. Introductions to these methods are provided in [20] and [7].

Further approaches include mirror Langevin methods, proximal samplers, and non-reversible MCMC algorithms based on piecewise deterministic Markov processes. Their convergence guarantees depend on the target distribution, the available information, and the distance or divergence used to measure error. For non-log-concave targets, obtaining useful quantitative bounds can be challenging, particularly when PI or LSI estimates are unavailable or have unfavourable constants. Nevertheless, these difficulties do not rule out efficient sampling for every such target.

Diffusion models provide an alternative approach under a different information setting: samples from the target are available for learning time-dependent score functions. As discussed in the next chapter, [5, 19, 4] establish polynomial sampling guarantees under assumptions on score-estimation error, the forward process, and numerical approximation. These guarantees concern the sampling procedure conditional on sufficiently accurate scores; they do not automatically include the cost of obtaining training data or learning those scores.

4 Diffusion Models

Diffusion Models (DMs) are deep generative models exhibiting state-of-the-art performances in many applications, e.g. image synthesis, video generation and molecule design. They combine a fascinating theoretical foundation, such as the theory of diffusion processes, with an increasing interest from researchers due to its applicability in many businesses. As a consequence of this there has been a significant growth in the amount of research papers published in this field; thus, it is quite difficult to stay abreast with the continuous developments made by the community. However, despite this, there still is a lot of work to do toward both theoretical and empirical directions.

These models operate by defining a forward-in-time SDE that progressively destroys the structure of the data by injecting noise, ultimately transforming a sample into pure noise. Under certain regularity conditions, there exists a corresponding reverse-in-time SDE that can reconstruct the data structure, generating new samples from noise and thus enabling generative modeling. However, some approximations are necessary, so, in practice, we sample from a distribution that is as close as possible to p_{data} .

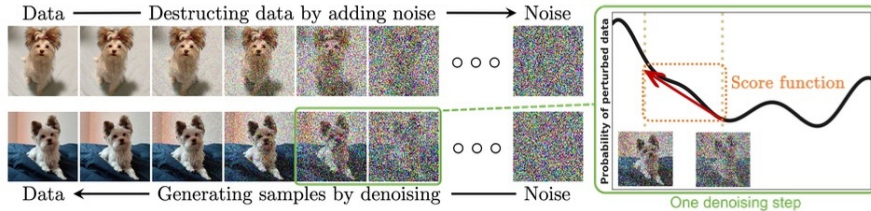


Figure 2: Generation of a new sample (on the left); illustration of score function role in one denoising step (on the right).

4.1 Overview

Usually, the forward SDE, whose role is to destroy the data structure, is given by the Ornstein-Uhlenbeck (OU) process $\{X_t^x; t \geq 0, x \in \mathbb{R}^d\}$, which is governed by the equation,

$$dX_t = -X_t dt + \sqrt{2} dB_t, \quad X_0 = x, \quad (4.1)$$

where $(B_t)_{t \geq 0}$ is a standard d -dimensional Brownian Motion and the initial value is a sample from the data distribution, $X_0 \sim p_{data}$. This is the unique time-homogeneous

Markov process which is also a Gaussian process. Furthermore, it is an instance of the Langevin diffusion with unique and reversible stationary distribution the standard d -dimensional Gaussian, $d\gamma^d(x) = (2\pi)^{-d/2}e^{-|x|^2/2}dx$. Its generator is $\mathcal{L} = \Delta - x \cdot \nabla$ and it is self-adjoint in the $\mathbb{L}^2(\gamma^d)$ -space. The OU process is one of the few examples of stochastic processes having a closed-form solution; in this case, it is given by,

$$X_t^x = e^{-t} \left(x + \sqrt{2} \int_0^t e^s dB_s \right),$$

where we have the Itô stochastic integral $\int_0^t e^s dB_s \sim \mathcal{N} \left(0, \int_0^t e^{2s} ds I_d \right)$, with the variance given by the Itô isometry. Therefore, $X_t^x \sim \mathcal{N} \left(e^{-t}x, (1 - e^{-2t})I_d \right)$, so that the semigroup $(P_t)_{t \geq 0}$ has an explicit representation, given by,

$$P_t f(x) = \mathbb{E} (f(X_t^x)) = \int_{\mathbb{R}^d} f \left(e^{-t}x + \sqrt{1 - e^{-2t}}y \right) d\gamma^d(y), \quad t \geq 0, x \in \mathbb{R}^d.$$

It is standard knowledge that the OU process (better, the standard Gaussian) satisfies both the log-Sobolev and the Poincaré inequalities with constants equal to 1.

The stationary measure of the forward process is called, in the context of DMs, prior distribution because the generative procedure starts by sampling from it; sampling from the d -dimensional standard Gaussian is easy, that is why the OU process has been, up to now, the default choice. As shown in [1] through a rearrangement of the Fokker-Planck, equation (4.1) admits a reverse-time SDE which has the following expression,

$$d\overleftarrow{X}_t = \left\{ \overleftarrow{X}_t + 2\nabla_{\overleftarrow{x}_t} \ln p_{T-t}(\overleftarrow{X}_t) \right\} dt + \sqrt{2}dB_t, \quad \overleftarrow{X}_0 \sim p_T, \quad (4.2)$$

where $\overleftarrow{X}_t := X_{T-t}$ for $t \in [0, T]$ is the backward diffusion process. Note that the two processes share the same marginals distributions. The *score function* or *Stein score*, $\nabla \ln p_t$, is a vector field which points towards regions of the space where the high-dimensional vector $\overleftarrow{X}_t$ has higher density, namely towards the modes of the distribution, as Langevin dynamics does. As derived in [28], the reverse-SDE (4.2) is associated with a corresponding ODE, called *Probability Flow ODE*, which has the same marginal

distributions along the trajectory and has the following dynamics,

$$\dot{X}_t = X_{T-t} + \nabla \ln p_{T-t}(X_{T-t}). \quad (4.3)$$

Therefore, this is an alternative way of carrying out the generation of new samples (through ODE solvers). However, the absence of stochastic noise might prevent it to discover low probability regions of distributions.

The denoising score-matching objective introduced by [14, 31] is

$$\min_{\theta} \mathbb{E}_{t \sim U[0, T]} \left\{ \lambda(t) \mathbb{E}_{X_0 \sim p_{data}} \mathbb{E}_{X_t | X_0 \sim p_{t|0}} [\|s_{\theta}(X_t, t) - \nabla_{x_t} \ln p(X_t | X_0)\|^2] \right\}, \quad (4.4)$$

where $p_{t|0}$ is the conditional density of X_t given X_0 , and $\lambda(t)$ is a positive weighting function.

However, not knowing the data distribution we cannot marginalize the conditional distribution of the OU process; therefore, we need to estimate it using a deep neural network, thus introducing an approximation error. The loss function used in the training procedure is given by equation (4.4); in practice, that loss is replaced with the empirical counterpart that uses the set of i.i.d. samples from p_{data} , namely $(X_0^{(1)}, \dots, X_0^{(n)}) \stackrel{iid}{\sim} p_{data}$. Then, a set of score functions is estimated; in fact, let $h > 0$ be the step-size of the discretization, so that the time interval $[0, T]$ is divided into N sub-intervals indexed by $k \in \{0, \dots, N\}$ such that $T = Nh$. Let's assume that during the training procedure we have trained a deep neural network to estimate the score function, $s_{\theta}(\overleftarrow{X}_{kh}, T - kh) \approx \nabla \ln p_{T-kh}$, for each time $k \in \{0, \dots, N\}$. The treatment of Noise Conditional Score Networks can be found in [27]. Then, between the discretization steps, $t \in [kh, (k+1)h]$, we freeze the score, so that we obtain the following SDE,

$$d\overleftarrow{X}_t = \left[\overleftarrow{X}_t + 2s_{\theta}(\overleftarrow{X}_{kh}, T - kh) \right] dt + \sqrt{2}dB_t, \quad t \in [kh, (k+1)h], \quad (4.5)$$

which represents the so-called exponential integrator scheme (see [34]). This is again a linear SDE, thus it has a closed form solution, namely $\overleftarrow{X}_{(k+1)h} | \overleftarrow{X}_{kh}$ is normally distributed. We just introduced the second approximation of the algorithm, the discretization scheme. Finally, in order to start the sampling procedure we sample from the standard d-dimensional Gaussian, γ^d , because we do not know the marginal p_T which would be

available if we knew p_{data} . Fortunately, the process (4.1) converges exponentially fast, that is $\text{KL}(p_T || \gamma^d) \leq e^{-2T} \text{KL}(p_{data} || \gamma^d)$, as a result of the LSI. Therefore we introduce a third approximation in our algorithm, $p_T \approx \gamma^d$. Then, denote with $q_t := \text{law}(\overleftarrow{X}_t)$ the law of the algorithm at time t .

4.2 Convergence

It is of great interest to study theoretical convergence guarantees of such models, that is, given a set of assumptions (e.g. finite second moment of p_{data}) about the model, state upper bounds on the distance between the measure we sample from, q_T , and the target one, p_{data} , as a function of the approximations that were implemented and of the assumptions made.

Most of the times the distributions from which we want to generate new sample are really high-dimensional (e.g. images can have millions of dimensions), probably they have several modes and are non log-concave. This prevents the assumption that p_{data} satisfies a LSI, as in [18]. There are also some papers which present results (bounds) that scale exponentially in the dimension of the data and some parameters, as in [11]. One of the main problems is whether the data distribution is absolutely continuous with respect to the Lebesgue measure in $\mathbb{R}^d$ or if, instead, it satisfies the manifold hypothesis, that is, p_{data} is supported on a lower-dimensional manifold embedded in the d -dimensional euclidean space. This might sound like a reasonable hypothesis since, as in the case of images, it is quite unlikely that we need millions of dimensions to characterize a single sample. If you want to know how to estimate the intrinsic data dimension take a look at [29]. As a consequence of the acceptance of the manifold hypothesis we encounter several problems: the score function is not well-defined and will explode when $t \downarrow 0$ due to the fact that $\nabla \log p_t = \nabla p_t / p_t$ with the denominator being zero outside of the manifold. Another problem is that $\text{KL}(p_{data} || \gamma^d) = \infty$ if $p_{data} \ll \gamma^d$ does not hold, so that we can not describe the convergence of the forward process to equilibrium. However, polynomial guarantees have been obtained also under the manifold hypothesis.

Let's consider the results in [5], with similar results obtained in [19] and [4]. They make three assumptions about the data distribution, p_{data} .

1. The score function is L -Lipschitz for each $t \geq 0$:

$$\|\nabla \ln p_t(x) - \nabla \ln p_t(y)\| \leq L\|x - y\|, \quad \forall x, y \in \mathbb{R}^d. \quad (4.6)$$

2. The data distribution has finite second moment, $m_2^2 := \mathbb{E}_{p_{data}}[\|\cdot\|^2] < \infty$.
3. For all $k = 1, \dots, N$ we have an L^2 -bound on the score estimation error,

$$\mathbb{E}_{p_{kh}}[\|s_\theta(\cdot, kh) - \nabla \ln p_{kh}(\cdot)\|^2] \leq \epsilon_{score}^2. \quad (4.7)$$

The first assumption assures us that $p_{data} \ll dx$ so that it does not lie on a submanifold; this is crucial in order to state the first theorem. The second, requiring finite variance, is a quite standard assumption in order to guarantee a well-posed model. The last assumption is crucial for the results of this paper, given the statement there reported: *"Given an L^2 -accurate score estimate, diffusion models can sample from (essentially) any data distribution."* Deep neural networks take in input high-dimensional data and are parametrized by an impressive number of weights and biases; furthermore, the nonlinearities introduced by the so-called activation functions make the loss landscape non-convex and highly complex, so that the optimization procedure is mostly empirical and not theoretically understandable yet. Consequently, it is not possible to give very precise guarantees for the estimation error. Therefore, they reduced the problem of sampling from a high-dimensional multimodal non log-concave distribution to the task of being able to estimate with L^2 -accuracy the score functions, $\nabla \ln p_t$. This requirement is more reasonable than the one of L^∞ -accuracy introduced in previous works due to the fact that the neural network might not be able to generalize super-well in low density regions of the distributions because we might have few samples, coming from those regions, for the training procedure. However, empirically, these architectures seem to be quite good in estimating the score functions.

The forward and reverse SDEs are defined as in (4.1) and (4.2). We can state the main theorem of the paper.

Theorem 11. *Suppose that assumptions 1, 2 and 3 hold. Let q_T be the output of equation (4.5) started with a sample from γ^d , and suppose that the step size $h := T/N$*

satisfies $h \lesssim 1/L$ where $L \geq 1$. Then we have the following bound

$$\|q_T - p_{data}\|_{TV} \lesssim e^{-T} \sqrt{\text{KL}(p_{data} \|\gamma^d)} + (L\sqrt{dh} + Lm_2h)\sqrt{T} + \epsilon_{score}\sqrt{T}. \quad (4.8)$$

Then, assuming $\text{KL}(p_{data} \|\gamma^d) \leq \text{poly}(d)$ (which is standard assumption) and $m_2 \leq d$ (meaning that the distribution does not have high variance), choosing $T \asymp \log(\text{KL}(q \|\gamma^d)/\epsilon)$ and $h \asymp \epsilon^2/L^2d$, we have

$$\begin{aligned} \|q_T - p_{data}\|_{TV} &\leq \tilde{O}(\epsilon + \epsilon_{score}), \text{ in} \\ N &= \tilde{\Theta}\left(\frac{L^2d}{\epsilon^2}\right) \text{ iterations.} \end{aligned}$$

First of all, this computational complexity is very good given that it scales polynomially in all the parameters of interest. Therefore, assuming an L^2 -accurate score estimate (for each time along the discretization scheme), we can sample with polynomial complexity even when p_{data} is non log-concave, but satisfies the other two assumptions, i.e. it has full support and has finite second moment. This sampling guarantee matches state of the art mixing time of the LMC algorithm under a LSI assumption, as in [30] and [8]. The bound (4.8) has three terms on the right-hand side. The first one concerns the convergence of the forward SDE, $p_T \approx \gamma^d$; we know that the OU process with γ^d as stationary measure converges exponentially fast (at rate $\exp(-2t)$) in relative entropy as implied by the LSI. The second term is about the discretization error of the trajectory and the last term due to the score function error. Thus, these are three approximation used to bound the convergence of the model.

It turns out that the best way to study Markov diffusions is via relative entropy using the Girsanov's theorem to bound the divergence between path measures. Let's give a sketch of the proof, but first some notation:

- The data distribution is $p_{data} = p_0$.
- The forward process is denoted $(X_t)_{t \in [0, T]}$, and $X_t \sim p_t$.
- The reverse process is denoted $(\bar{X}_t)_{t \in [0, T]}$, where $\bar{X}_t := X_{T-t} \sim p_{T-t}$.
- The sampling algorithm is denoted $(\hat{X}_t)_{t \in [0, T]}$, where $\hat{X}_t \sim q_t$. Recall that we initialize at $q_0 = \gamma^d$, the standard Gaussian measure.

- The process $(\overleftarrow{X}_t^{p_T})_{t \in [0, T]}$, where is the same as $(\overleftarrow{X}_t)_{t \in [0, T]}$, except that we initialize this process at p_T rather than at γ^d . We write $\overleftarrow{X}_t^{p_T} \sim q_t^{p_T}$.

It is useful to consider a single stochastic process, $(X_t)_{t \in [0, T]}$, and two different measures over the path space

$$\mathcal{C}([0, T]; \mathbb{R}^d)$$

. The considered path measures are:

- $\overleftarrow{P}_T$, under which $(X_t)_{t \in [0, T]}$ has the law of the reverse process (4.2);
- $Q_T^{p_T}$, under which $(X_t)_{t \in [0, T]}$ has the law of the SGM algorithm initialized at p_T (corresponding to the process $(\overleftarrow{X}_t^{p_T})_{t \in [0, T]}$).

Let's proceed with the sketch of the proof.

Sketch of Proof of Theorem 11. Let's recall a consequence of Girsanov's theorem.

Theorem 12. For $t \in [0, T]$, let $\mathcal{L}_t = \int_0^t b_s dB_s$ where B is a P -Brownian motion. Assume $\mathbb{E}_P \left[\int_0^T \|b_s\|^2 ds \right] < \infty$. Then, $\mathcal{L}$ is a P -martingale in the Hilbert space $\mathbb{L}^2(P)$. Then, if

$$\mathbb{E}_P \mathcal{E}(\mathcal{L})_T = 1, \quad \text{where} \quad \mathcal{E}(\mathcal{L})_t := \exp \left(\int_0^t b_s dB_s - \frac{1}{2} \int_0^t \|b_s\|^2 ds \right), \quad (4.9)$$

then $\mathcal{E}(\mathcal{L})$ is also a P -martingale and the process

$$t \mapsto B_t - \int_0^t b_s ds \quad (4.10)$$

is a Brownian motion under $Q := \mathcal{E}(\mathcal{L})_T P$, the probability distribution with density $\mathcal{E}(\mathcal{L})_T$ w.r.t. P .

The condition (4.9) is needed in order to apply the Girsanov's theorem. If it holds, then we can apply the theorem to $P = \overleftarrow{P}_T$ and

$$b_t = \sqrt{2} (s_\theta(X_{kh}, T - kh) - \nabla \ln p_{T-t}(X_t)), \quad (4.11)$$

where $t \in [kh, (k+1)h]$. This tells us that under $Q = \mathcal{E}(\mathcal{L})_T \overleftarrow{P}_T$, there exists a Brownian

motion $(\tilde{B}_t)_{t \in [0, T]}$ such that

$$dB_t = \sqrt{2} (s_\theta(X_{kh}, T - kh) - \nabla \ln p_{T-t}(X_t)) dt + d\tilde{B}_t. \quad (4.12)$$

Recall that under $\overleftarrow{P}_T$ we have a.s.

$$dX_t = \{X_t + 2 \nabla \ln p_{T-t}(X_t)\} dt + \sqrt{2} d\tilde{B}_t, \quad X_0 \sim p_T. \quad (4.13)$$

The equation above still holds Q -a.s. since $P \ll \overleftarrow{P}_T$ (even if B is no longer a Q -Brownian motion). Substituting (4.12) into (4.13) we have Q -a.s.,

$$dX_t = \{X_t + 2 s_\theta(X_{kh}, T - kh)\} dt + \sqrt{2} d\tilde{B}_t, \quad X_0 \sim p_T. \quad (4.14)$$

This translates in the fact that, under Q , the distribution of X is the sampling algorithm started at p_T , i.e., $Q = Q_T^{p_T} = \mathcal{E}(\mathcal{L})_T \overleftarrow{P}_T$. Therefore,

$$\text{KL}(\overleftarrow{P}_T \| Q_T^{p_T}) = \mathbb{E}_{\overleftarrow{P}_T} \ln \frac{d\overleftarrow{P}_T}{dQ_T^{p_T}} = \mathbb{E}_{\overleftarrow{P}_T} \ln \mathcal{E}(\mathcal{L})_T^{-1} \quad (4.15)$$

$$= \sum_{k=0}^{N-1} \mathbb{E}_{\overleftarrow{P}_T} \int_{kh}^{(k+1)h} \|s_\theta(X_{kh}, T - kh) - \nabla \ln p_{T-t}(X_t)\|^2 dt, \quad (4.16)$$

where we applied the martingale equality $\mathbb{E}_{\overleftarrow{P}_T} \mathcal{L}_t = 0$.

The equality (4.15) allows us to bound the relative entropy between the reverse process and sampling algorithm started at the same distribution, p_T .

Then, in order to conclude the proof we need to prove that

$$\|Q_T^{p_T} - \overleftarrow{P}_T\|_{TV}^2 \leq \text{KL}(\overleftarrow{P}_T \| Q_T^{p_T}) \lesssim (\epsilon_{score}^2 + L^2 dh + L^2 m_2^2 h^2) T, \quad (4.17)$$

where the first inequality is the Pinsker-Csizsar- Kullback inequality.

Finally, by data processing inequality,

$$\|q_T - p_{data}\|_{TV} \leq \|Q_T - Q_T^{p_T}\|_{TV} + \|Q_T^{p_T} - \overleftarrow{P}_T\|_{TV} \leq \|p_T - \gamma^d\|_{TV} + \|Q_T^{p_T} - \overleftarrow{P}_T\|_{TV}, \quad (4.18)$$

where the first term on the right-hand-side is bounded by the convergence of the forward OU process while the second by (4.17), so to obtain

$$\|q_T - p_{data}\|_{TV} \lesssim e^{-T} \sqrt{\text{KL}(p_{data} \|\gamma^d)} + (L\sqrt{dh} + Lm_2h)\sqrt{T} + \epsilon_{score}\sqrt{T}. \quad (4.19)$$

□

Notwithstanding, we can obtain similar results if we assume that the manifold hypothesis holds true and that p_{data} has bounded support; in this case, the new target becomes p_t , $t > 0$ so that the probability distribution from which we effectively sample is q_{T-t} . Then, it can be shown that $\|q_{T-t} - p_t\|_{TV} \leq \epsilon_{TV}$ and $\mathcal{W}_2(q_{T-t}, p_{data}) \leq \epsilon_{\mathcal{W}_2}$ given specific choices of the parameters. The early stopping technique is quite used to sample from a distribution having full support, in order to not have an exploding behavior with the Stein score at time $t \approx 0$.

5 Conclusions

This work has presented a theoretical study of the convergence properties of stochastic processes used in sampling and generative modelling, covering Markov processes, classical sampling algorithms, and diffusion models. We have examined how functional inequalities, ergodicity, and spectral properties provide rigorous guarantees for convergence to a target distribution. The analysis of classical sampling algorithms has highlighted the trade-offs between computational cost and sampling accuracy, particularly in high-dimensional settings. The discussion of diffusion models has shown how reversing a forward diffusion enables generative modelling, while score estimation, time discretisation, and approximate initialisation introduce errors that must be accounted for in convergence bounds. Despite these theoretical developments, important challenges remain. In particular, sampling guarantees based on accurate score estimates must be connected to the statistical and computational performance of neural-network training. The manifold hypothesis also raises questions about score regularity and the appropriate choice of convergence metric. Future research could further clarify these connections, refine existing bounds, and develop training and sampling methods whose guarantees remain informative under realistic assumptions.

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A Technical appendix

A.1 Convexity

For this subsection of the appendix we will mainly follow the review from [26].

Definition 12. A function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex if, for all $\theta \in [0, 1]$, it satisfies the following inequality

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y), \quad \forall x, y \in \mathbb{R}^d.$$

This definition of convexity is the most basic one and doesn't make any assumption about the curvature of the function. The following one is a stronger notion of convexity which, therefore, applies to a subset of the set of convex functions.

Definition 13. A function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is α -strongly convex if, for $\alpha > 0$ and all $\theta \in [0, 1]$, we have

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y) - \frac{1}{2}\alpha\theta(1 - \theta)\|x - y\|^2, \quad \forall x, y \in \mathbb{R}^d.$$

Equivalently, it holds that $f(x) - \frac{1}{2}\alpha\|x\|^2$ is convex. The strong convexity assumption implies a lower bound for the function's curvature; in fact, if f is differentiable then the above inequality becomes $f(y) \geq f(x) + \nabla f(x)^T(y - x) + \frac{1}{2}\alpha\|y - x\|^2$, where we can notice that the function will have positive curvature, meaning that the eigenvalues of the Hessian (if well defined) will be lower bounded by α , which can be written in terms of positive semi-definiteness as,

$$\nabla^2 f(x) \succeq \alpha I \iff \nabla^2 f(x) - \alpha I \succeq 0.$$

Furthermore, a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is β -smooth if its gradient is Lipschitz continuous with constant β , i.e.,

$$\|\nabla f(x) - \nabla f(y)\| \leq \beta\|x - y\|, \quad \forall x, y \in \mathbb{R}^d.$$

Consequently, this is the same as requiring that the largest eigenvalue of $\nabla^2 f$ is at most equal to β .

Now we can define the notion of log-concave distribution.

Definition 14. A density function π with respect to the Lebesgue measure λ on the measurable space $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ is log-concave if $\pi = e^{-\psi}$, where $\psi : \mathbb{R}^d \rightarrow \mathbb{R}$ is a convex function.

If we didn't have access to the normalizing constant then we may express it as $\pi \propto e^{-f}$, where f is convex.

Consequently, we have the following definition for a Probability Measure (PM).

Definition 15. A probability measure P on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ such that the affine hull of $\text{supp}(P)$ has dimension d is a log-concave measure if and only if it has a log-concave density function $\pi = e^{-\psi}$ on $\mathbb{R}^d$; that is, we have

$$P(A) = \int_A \pi d\lambda, \quad A \in \mathcal{B}(\mathbb{R}^d).$$

We can go on by defining the strong log-concavity of a density function π on $\mathbb{R}^d$.

Definition 16. Let Σ be a $d \times d$ positive definite matrix and let $\mu \in \mathbb{R}^d$. Then a random vector X and its density π are strongly log-concave, and write $\pi \in SLC_2(\mu, \Sigma, d)$, if

$$\pi(x) = g(x) \gamma_{\Sigma}(x - \mu), \quad x \in \mathbb{R}^d,$$

where g is a log-concave function and $\gamma_{\Sigma}(\cdot - \mu)$ denotes the density of a d -dimensional Gaussian.

Let's notice that for the special case identified by $\Sigma = \sigma^2 I$ we have that the negative log-likelihood, namely $-\log \pi(x)$, are strongly convex with $\alpha = 1/\sigma^2$; if we think about a Gaussian we know that as the variance increases the density becomes less concentrated with fatter tails. Between log-concave distributions and strongly log-concave ones there exists an intermediate class: log-concave distributions of order p .

Definition 17. A log-concave distribution of order $p \geq 1$ is identified by a density function of the form $\pi(x) = x^{p-1} g(x)$, $x > 0$, where g is log-concave on $\mathbb{R}_+$.

It can be shown that log-concave distributions of order $p > 1$ are more concentrated than simple log-concave ones. Examples of such distributions are: the Gamma with $\alpha \geq 1$ degrees of freedom is log-concave of order α , while the Beta with parameter $\alpha \geq 1$ and $\beta \geq 1$ is still log-concave of order α .

A.2 Spectral Theory

In linear algebra particular emphasis is given to the spectral theorem, which is an instance of the Singular Value Decomposition. The theorem states that, if we have an Hermitian matrix A (or a symmetric matrix), then we can diagonalize it through a change of basis. Each element on the diagonal represents a real eigenvalue of A ; the new basis is an orthonormal basis of eigenvectors of A . Thus, A can be written as a linear combination of pairwise orthogonal projections, called its spectral decomposition. Formally, given a symmetric matrix A , representing a linear map $f : V \rightarrow V$, with eigenvalues $\lambda_1, \dots, \lambda_n$ and corresponding eigenvectors $v_1, \dots, v_n$, we can write it as $A = \sum_{i=1}^n \lambda_i E_i$, where $E_i = v_i v_i^T$, with $\|v_i\| = 1 \ \forall i$, is the projection matrix onto the eigenspace associated with λ_i ; these eigenspaces are orthogonal, thus, we can decompose the vector space V on which the matrix A acts as $V = \oplus_{i=1}^n V_i$, where $V_i = \{v \in V : Av = \lambda_i v\}$. The properties of such projections matrices are the following: $E_i E_j = 0 \ \forall i \neq j$ due to orthogonality, $E_i^2 = E_i$ and $\sum_{i=1}^n E_i = Id$.

Therefore, when we apply a symmetric matrix A to some vector $v \in V$, we can decompose its effect as a linear combination of the effects of the single eigenvalues. Furthermore, diagonalizing a matrix offers several advantages, e.g. computing A^k , e^A . When moving to an infinite-dimensional setting, the theory gets a little bit more intricate. However, the general idea is the same. Instead of talking about diagonalization we talk about spectral decomposition.

Definition 18. *Given a linear operator A on a Banach space $\mathcal{B}$, the resolvent set $\rho(A)$ of A is the set of all complex values $\lambda \in \mathbb{C}$ such that the range of $\lambda Id - A$ is dense in $\mathcal{B}$ and such that $\lambda Id - A$ has a bounded inverse $R_\lambda = (\lambda Id - A)^{-1}$, called the resolvent operator. Then the spectrum $\sigma(A)$ of A is $\mathbb{C} \setminus \rho(A)$.*

This spectrum is made by three components: the *point spectrum*, which contains the eigenvalues of A , namely those values of λ such that there exists a non-zero solution $f \in \mathcal{B}$ to $Af = \lambda f$, where f is an eigenfunction; the *continuous spectrum*, made by those values of λ for which the range of $\lambda Id - A$ is dense in $\mathcal{B}$ but the inverse is not bounded; the *residual spectrum* which contains the values of λ which do not satisfy neither of the above conditions. It turns out that for self-adjoint operators on separable Hilbert spaces the spectrum, namely the values of λ , is always real and the residual spectrum is empty.

How do we represent the spectral decomposition of such an operator? Given a positive self-adjoint operator A on a real, separable Hilbert space $\mathcal{H}$, a spectral decomposition is an increasing family of closed linear subspaces of $\mathcal{H}$, namely $(\mathcal{H}_\lambda)_{\lambda \geq 0}$. This family is required to be right-continuous, i.e. $\bigcap_{\lambda' > \lambda} \mathcal{H}_{\lambda'} = \mathcal{H}_\lambda$ and $\bigcup_{\lambda \geq 0} \mathcal{H}_\lambda$ is dense in $\mathcal{H}$. Then, as we did in the finite-dimensional case, for every $\lambda \geq 0$ we consider the orthogonal projection P_λ onto $\mathcal{H}_\lambda$. The finite sum becomes an integral, as shown by the following theorem.

Theorem 13. *Given a positive self-adjoint operator A on $\mathcal{H}$, there exists a spectral decomposition $(E_\lambda)_{\lambda \geq 0}$ such that*

$$A = \int_0^\infty \lambda dE_\lambda.$$

Furthermore, for any measurable function ψ on $\mathbb{R}_+$, one may define the operator $\psi(A)$ as

$$\psi(A) = \int_0^\infty \psi(\lambda) dE_\lambda,$$

on functions $f \in \mathcal{H}$ such that $\|\psi(A)f\|^2 = \int_0^\infty \psi(\lambda)^2 d\langle E_\lambda f, f \rangle < \infty$, where we have the notation $\|E_\lambda f\|^2 = \langle E_\lambda f, f \rangle$.

It is quite clear that this is just a generalization of the formula $A = \sum_{i=1}^n \lambda_i E_i$. The range of integration is positive due to the fact that a positive operator has positive spectrum.

How does this apply to diffusion processes we work with? Well, in our case the infinitesimal generator, $\mathcal{L}$, is a negative operator; thus, its opposite, namely $-\mathcal{L}$, is a positive self-adjoint operator on the Hilbert space $\mathbb{L}^2(\mu)$. Its spectral decomposition is $-\mathcal{L} = \int_0^\infty \lambda dE_\lambda$. As a consequence of the previous theorem we have that $P_t f = \int_0^\infty e^{-\lambda t} dE_\lambda f$, for every $f \in \mathbb{L}^2(\mu)$ and $t \geq 0$. Finally, note that if the spectrum of A is discrete, namely made only by normal eigenvalues having finite-dimensional eigenspaces, then we can express the Hilbert space as $\mathbb{L}^2(\mu) = \bigoplus_{k \in \mathbb{N}} E_k$, where $(E_k)_k$ is a sequence of closed subspaces of $\mathbb{L}^2(\mu)$ such that E_k is the eigenspace associated with $-\lambda_k$, $\lambda_k \geq 0$. Then we decompose $f \in \mathbb{L}^2(\mu)$ as $f = \sum_{k \in \mathbb{N}} f_k$ and $-\mathcal{L}f = \sum_{k \in \mathbb{N}} \lambda_k f_k$, $P_t f = \sum_{k \in \mathbb{N}} e^{-\lambda_k t} f_k$.

A.3 Stochastic Calculus

Stochastic processes in continuous-time are often defined on infinite-dimensional spaces. For example, the canonical definition of the Brownian Motion, $(B_t)_{t \geq 0}$ with $B_t \in \mathbb{R}^d$, is on the probability space $(\Omega, \mathcal{F}, \mathbb{P}) = (\mathcal{C}(\mathbb{R}_+; \mathbb{R}^d), \mathcal{B}(\mathcal{C}(\mathbb{R}_+; \mathbb{R}^d)), \mathbb{W})$, where $\mathbb{W}$ is the Wiener measure, a probability measure on the path space of continuous trajectories $\mathcal{C}(\mathbb{R}_+; \mathbb{R}^d)$, so that

$$\mathbb{W}(A) = \mathbb{P}(t \mapsto B_t(\omega) \in A), \quad \forall A \in \mathcal{B}(\mathcal{C}(\mathbb{R}_+; \mathbb{R}^d)).$$

Then, we specialize the measurable sets to cylinder sets because they allow us to correctly define the Wiener measure via the finite-dimensional distributions of Brownian motion on such sets, thanks to the Kolmogorov extension theorem.

Girsanov's theorem enables us to define the Radon-Nikodym derivative $d\mathbb{P}/d\mathbb{Q}$ between two path measures so that if we change the measure we obtain a new process with different drift. Let's state the theorem.

Theorem 14. *Let $(B_t)_{t \in [0, T]}$ be a (standard) Brownian Motion under the Wiener measure $\mathbb{W}$ and let $(b_t)_{t \in [0, T]}$ be a progressive process with finite variance, $\mathbb{E}^{\mathbb{W}} \left(\int_0^T \|b_s\|^2 ds \right) < \infty$. Let $M_t := \int_0^t \langle b_s, dB_s \rangle$ for $t \in [0, T]$ denote a continuous local martingale and let $[M, M]_t := \int_0^t \|b_s\|^2 ds$ denote its quadratic variation. Let's define the exponential martingale*

$$\mathcal{E}(M) := \exp \left(M - \frac{1}{2} [M, M]_t \right).$$

Assume that $\mathcal{E}(M)$ is a $\mathbb{W}$ -martingale and define the path measure $\mathbb{P}$ as

$$\frac{d\mathbb{P}}{d\mathbb{W}} = \mathcal{E}(M)_T.$$

Then, under $\mathbb{P}$ the map

$$t \mapsto \tilde{B}_t := B_t - [B, M]_t = B_t - \int_0^t b_s ds,$$

is a (standard) Brownian Motion.

Therefore, if we change the path measure to $\mathbb{P}$ we introduce the drift component

$$\int_0^t b_s \mathrm{d}s.$$